UNIT I – Metal Cutting

23A03501 MACHINING PROCESSES / B.Tech Mech (JNTUK) – Exam Notes

I. Elementary Theory of Metal Cutting

Metal cutting = removing unwanted material from a workpiece in the form of **chips** using a wedge-shaped, harder tool, to get required shape, size & finish.

Elements of cutting process

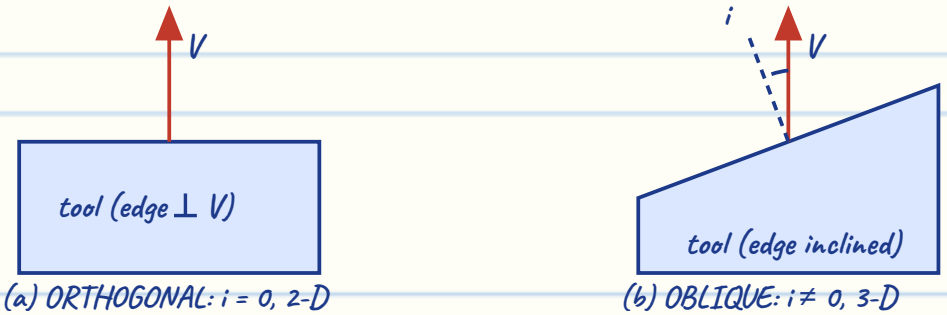
- **Workpiece** (softer) + **cutting tool** (harder wedge) + **machine**
- Relative motion tool/work: **primary (cutting) motion** → speed V
- **Secondary (feed) motion** → feed f; **depth of cut** d (infeed)
- Result: **chip** (by shearing in **shear zone**) + machined surface

Cutting parameters (turning)

- **Cutting speed** $V = \pi DN/1000$ m/min (D mm, N rpm)
- **Feed** f = mm/rev (lathe), mm/stroke (shaper), mm/tooth (milling)
- **Depth of cut** $d = (D_1 - D_2)/2$ mm, ⊥ to feed
- Uncut chip thickness $t = f \sin(\text{SCEA-complement})$; width $b = d / \sin\varphi_s$

$$MRR = 1000 \cdot V \cdot f \cdot d \ \ mm^3/min \quad | \quad \text{Machining time } Tm = L / (f \cdot N) \ min$$

Orthogonal vs Oblique cutting



(a) ORTHOGONAL: $i = 0$, 2-D (b) OBLIQUE: $i \neq 0$, 3-D

Fig 1. Orthogonal and oblique cutting (plan view); i = inclination angle

Orthogonal	Oblique
Edge ⊥ cutting velocity (i = 0)	Edge inclined at i to the normal
2 force components (Fc, Ft)	3 components (Fc, Ft, Fr)
Chip flows ⊥ edge, plane strain	Chip flows sideways, curls helically
Edge longer than chip width	Longer edge contact → better life, finish
Used for theory; e.g. parting, broaching	Practical turning, milling, drilling

2. Single Point Cutting Tool – Geometry

- **Shank** – held in tool post; **Flank** – surface below edge facing machined surface
- **Face (rake face)** – chip slides over it; **Heel** – lowest portion of tool
- **Nose** – junction of side & end cutting edges (radius r)
- **Side cutting edge** does main cutting; **end cutting edge** clears finished surface

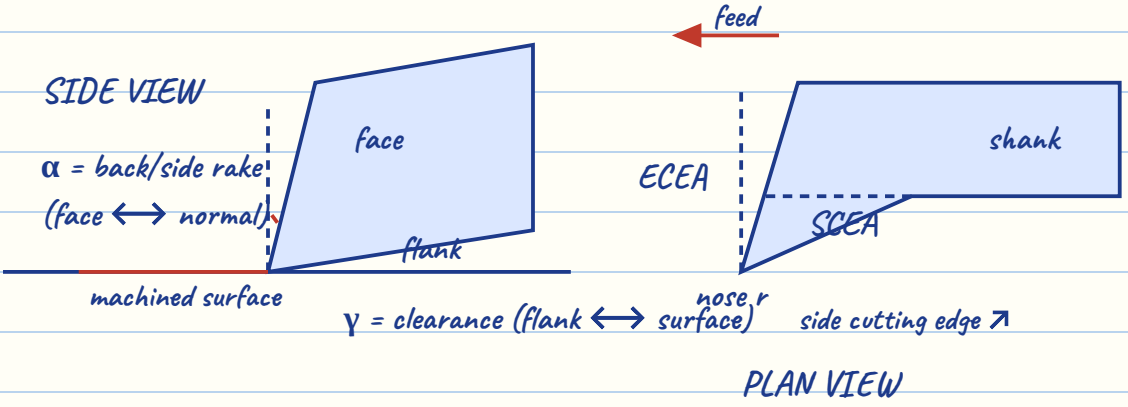


Fig 2. Single point tool – side and plan views with angles

Tool angles & functions

- **Back rake** αb (along shank), **side rake** αs (across) – chip flow, force
  - +ve rake: ↓ force, ↓ power, better finish, but weak edge (ductile, HSS)
  - –ve rake: strong edge, takes shocks (carbide/ceramic, hard work)
- **Relief (clearance)** end & side (5–10°): avoids rubbing on work
- **SCEA** Cs (15–30°): ↑ edge length, thins chip; **ECEA** Ce (5–15°): finish
- **Inclination angle** λ: directs chip (–ve λ → chip away from finished surface)
- **Nose radius** r: ↑ strength & finish; too big → chatter

Tool signature (designation)

System	Order of angles	Example
ASA (American, tool-in-hand)	αb – αs – end relief – side relief – ECEA – SCEA – r	8-14-6-6-6-15-1 (mm)
ORS (orthogonal rake, tool-in-use)	λ – αn – side relief γ – end relief γ_1 – ECEA φ_1 – cutting edge angle φ – r	10-8-6-6-6-75-1

- ORS principal cutting-edge (approach) angle $\varphi = 90° - SCEA(ASA)$
- Mnemonic ASA: **"Back Side End Side End Side Radius"** (3 rake/relief pairs)

3. Chip Formation

- Tool pushes into work → **compressive stress** → plastic deformation by **shear** along the **shear plane** (angle φ)
- Chip = material sheared, then slides on rake face (friction: **secondary deformation**)
- Chip is thicker ($t_c > t$) & shorter than uncut layer → **chip compression**

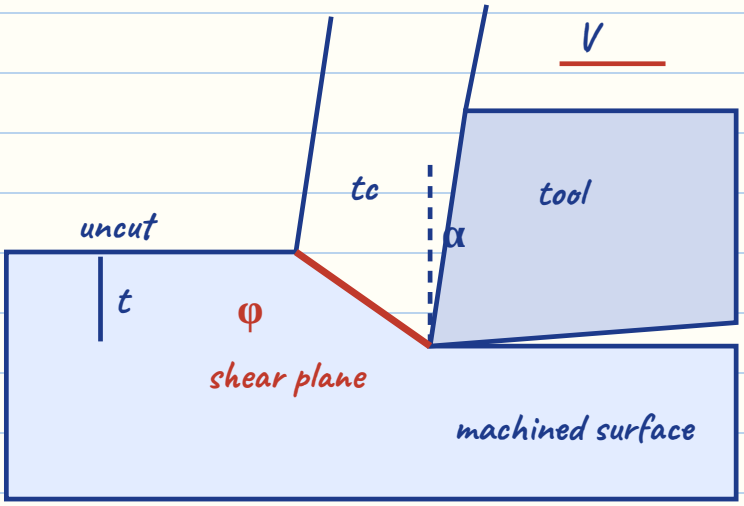


Fig 3. Orthogonal cutting model – shear plane, t, tc, φ, α

Chip thickness ratio & shear angle

$$r = t / t_c = \sin\varphi / \cos(\varphi - \alpha) \quad (r < 1)$$

$$\tan\varphi = r\cos\alpha / (1 - r\sin\alpha)$$

- Chip velocity $V_c = r \cdot V$; shear velocity $V_s = V\cos\alpha / \cos(\varphi - \alpha)$
- Shear strain $\gamma = \cot\varphi + \tan(\varphi - \alpha)$; also $r = V_c/V = L/L_c$ (length ratio)
- Large φ → thinner chip, less deformation, less force & heat (good)

★ Important

- Derivation: $AB = t/\sin\varphi = tc/\cos(\varphi-\alpha) \Rightarrow r = \sin\varphi / \cos(\varphi-\alpha)$
- Expand $\cos(\varphi-\alpha) \rightarrow r\cos\alpha + r\sin\alpha\tan\varphi = \tan\varphi \Rightarrow \tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$

4. Types of Chips

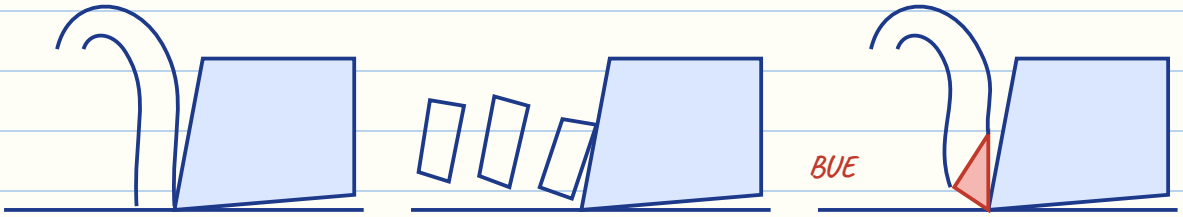


Fig 4. Types of chips

Type	Occurs when	Remarks
Continuous (ribbon)	Ductile metal, high V, small f, +ve rake, good coolant	Best finish, steady force; long chips = hazard → use chip breaker
Discontinuous (segmented)	Brittle (CI, bronze), low V, large f & d, −ve rake	Fluctuating force, rough finish, easy to handle
Continuous with BUE	Ductile, low–medium V, high friction	Poor finish, unstable edge, tool wear

- Also **serrated (shear-localised)** chips – Ti alloys, hard steels at high V

5. Built-Up Edge (BUE)

BUE = layer of work material **welded (pressure-weld) to the tool face** near the tip, strain-hardened, acting as a false cutting edge.

- **Causes:** high friction + high pressure + temperature, ductile work, low–medium V, no coolant/lubricant, small +ve rake, affinity of metals
- **Process:** builds up → becomes unstable → breaks off → carried by chip & work surface → repeats

Effects

- (−) rough, torn surface; poor dimensional accuracy (changes tool geometry)
- (−) tool crater/flank wear by fragments; force fluctuation, vibration
- (+) protects the tool edge, ↑ effective rake → ↓ force (at low speed, roughing only)

Remedies / how to eliminate

- ↑ cutting speed (above ~ recrystallisation temperature) · ↓ feed
- ↑ +ve rake angle · use sharp, polished tool; lower friction
- Effective **cutting fluid** (EP additive) · tool with low affinity (carbide/coated)

6. Chip Breakers

- Long continuous chips → danger to operator, tangle, damage surface, poor coolant access
- **Chip breaker** curls & fractures chip into short pieces (6-shaped / 'C' chips)

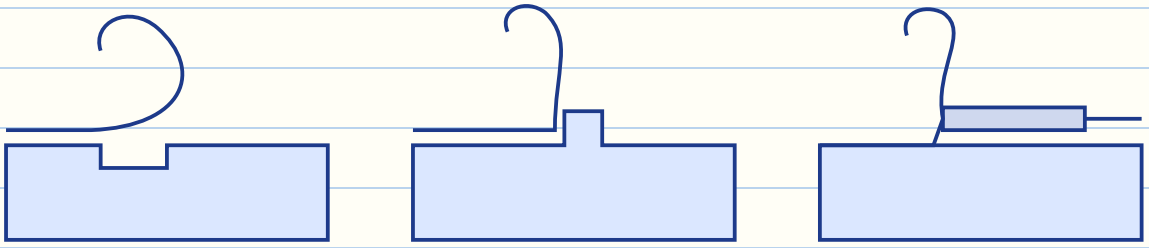


Fig 5. Chip breakers

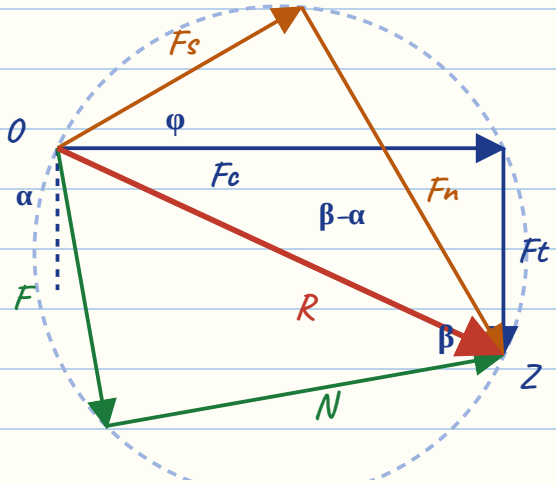
- **Groove type** – ground groove behind edge (width, depth, radius) – chip curls in groove
- **Obstruction type** – raised step/ridge ahead of edge deflects chip → bends → breaks
- **Clamp-on (plate) type** – adjustable carbide plate clamped on insert; common on indexable inserts
- Also by feed/speed variation, or **moulded insert geometry**; good break = f↑, d↑

7. Mechanics of Orthogonal Cutting – Merchant's Analysis

Assumptions

- Sharp tool, no rubbing on flank; 2-D (plane strain) orthogonal cutting
- Continuous chip, no BUE; shear takes place on single thin plane; uniform stress on it
- Constant V, constant t; tool rigid; chip in equilibrium under two equal-opposite forces

Merchant's force (circle) diagram



R = resultant (diameter)
Fc cutting force (power)
Ft thrust force (⊥ V)
F friction on rake face
N normal to rake face
Fs shear force (on plane)
Fn normal to shear plane
α rake, φ shear, β friction angle

Fig 6. Merchant's circle (R as diameter; each force pair makes a right angle on circle)

- Tool exerts R on chip (= R' by chip on tool): in equilibrium → R splits 3 ways
- (Fc, Ft) measured by dynamometer; (F, N) on rake face; (Fs, Fn) on shear plane
- Friction: **$\mu = F/N = \tan\beta$**, β = friction angle; $Ft/Fc = \tan(\beta - \alpha)$

Derived equations (from circle geometry)

$$F = Fc \sin\alpha + Ft \cos\alpha \qquad N = Fc \cos\alpha - Ft \sin\alpha$$

$$Fs = Fc \cos\varphi - Ft \sin\varphi \qquad Fn = Fc \sin\varphi + Ft \cos\varphi$$

$$\mu = F/N = (Ft + Fc \tan\alpha)/(Fc - Ft \tan\alpha) \;;\; \beta = \tan^{-1}\mu$$

$$R = Fc / \cos(\beta - \alpha) \;;\; Fs = R \cos(\varphi + \beta - \alpha)$$

- Shear stress $\tau s = Fs / As$, with $As = b \cdot t / \sin\varphi$ (shear plane area)
- Also $Fn = Fs \tan(\varphi + \beta - \alpha)$; $Fc = Fs \cos(\beta-\alpha)/\cos(\varphi+\beta-\alpha)$

Power & specific energy

$$Pc = Fc \cdot V \quad (W,\ V \text{ in } m/s) \;;\; u = Pc / MRR = Fc / (b \cdot t)$$

- Total energy = shear energy (Fs·Vs) + friction energy (F·Vc) → ~ 70–80 % in shear
- u (specific cutting energy): steel ~2–5, Al ~0.7–1, CI ~1–2 J/mm^3
- Size effect: u ↑ when t ↓ (finishing cuts need more energy per volume)
- Machine motor power = Pc / η ($\eta \approx 0.7$–0.85)

Merchant's shear angle relations

- **1st theory** – shear plane is where **energy (cutting force) is minimum**; τs constant: put $dFc/d\varphi = 0$

$$\varphi = 45° + \alpha/2 - \beta/2$$

- **2nd theory** – τ_s varies linearly with normal stress σ ($\tau_s = \tau_0 + k\sigma$):

$$2\varphi + \beta - \alpha = C \quad (C = \text{machining constant of material})$$

- Lee & Shaffer (slip-line): $\varphi = 45^\circ + \alpha - \beta$; Ernst–Merchant is the 1st form
- Meaning: ↑ rake α or ↓ friction β ⇒ ↑ φ ⇒ ↓ chip thickness, ↓ force (agrees with practice)
- Experimental values deviate for real metals – hence 2nd theory/C

★ *Important*

- Remember: $\varphi = 45 + \alpha/2 - \beta/2$; $2\varphi + \beta - \alpha = C$; $\mu = \tan\beta$
- $F_c \propto$ power; $F_t \propto$ deflection; $F/N = \mu$; $r = \sin\varphi/\cos(\varphi-\alpha)$

Solved Example 1 (Merchant's circle)

In orthogonal cutting: $t = 0.25$ mm, $t_c = 0.75$ mm, $\alpha = 10^\circ$, $F_c = 900$ N, $F_t = 450$ N, $V = 60$ m/min. Find φ, F_s, F_n, F, N, μ, power.

1. $r = 0.25/0.75 = 0.333$; $\tan\varphi = 0.333\cos10^\circ/(1 - 0.333\sin10^\circ) = 0.3485$ ⇒ **$\varphi = 19.2^\circ$**
2. $F_s = F_c\cos\varphi - F_t\sin\varphi = 900(0.944) - 450(0.329) =$ **702 N**
3. $F_n = F_c\sin\varphi + F_t\cos\varphi = 296 + 425 =$ **721 N**
4. $F = F_c\sin\alpha + F_t\cos\alpha = 156 + 443 =$ **600 N** ; $N = F_c\cos\alpha - F_t\sin\alpha = 886 - 78 =$ **808 N**
5. $\mu = F/N = 0.742 \Rightarrow \beta = 36.6^\circ$ (check: $F_t = F_c\tan(\beta-\alpha) = 900\tan26.6^\circ = 450$ ✓)
6. Power $P_c = F_c \cdot V = 900 \times 1$ m/s = **900 W**

8. Heat Generation in Metal Cutting

- Nearly all cutting work → **heat**; raises temperature of tool, chip, work

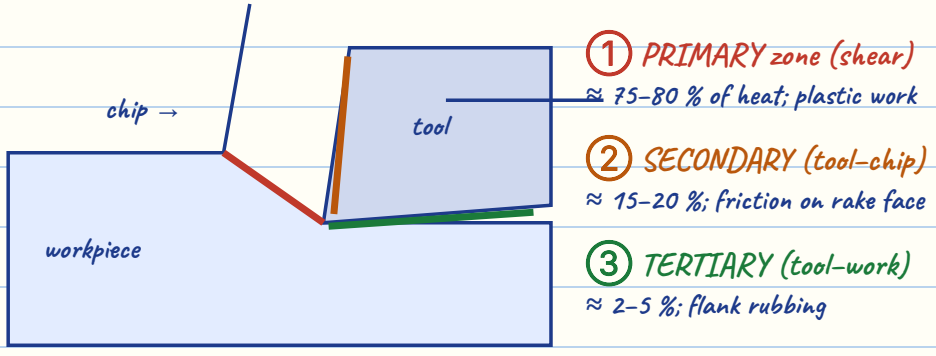


Fig 7. Heat zones in orthogonal cutting

- **Heat carried away**: chip ≈ 75–80 % · work ≈ 10–15 % · tool ≈ 5–10 % (at normal V)

- Max temperature on rake face ≈ at a distance behind edge (end of tool–chip contact)
- Temp. ↑ with V (strongest), f, d (weakly); $\theta \propto V^{0.4}$ (approx.)
- Effects: thermal softening of tool → **wear, BUE, crater**; work thermal expansion → inaccuracy; residual stress
- Control: coolant, sharp +ve rake tool, lower V, tool with low conductivity loss (carbide)
- Temperature measured by: tool–work thermocouple, embedded thermocouple, IR camera, calorimetry

9. Tool Wear & Tool Life

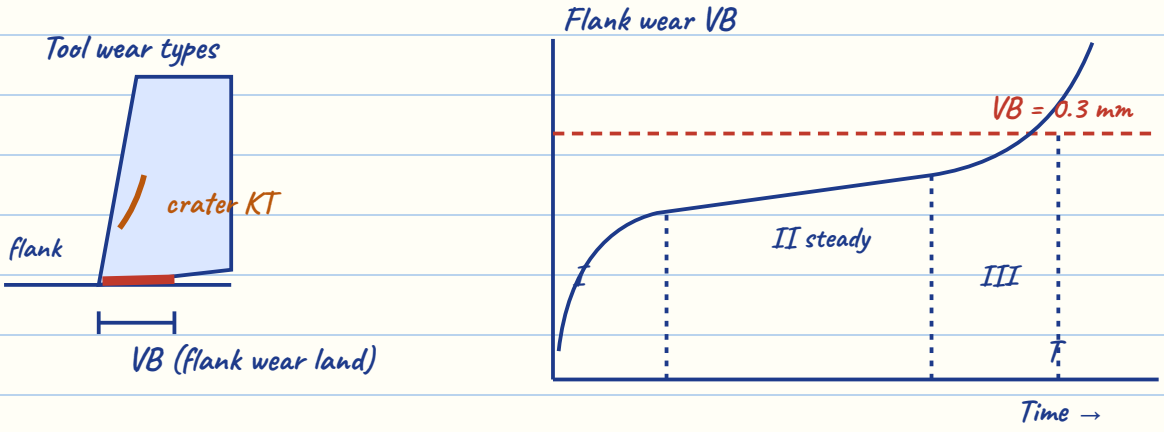


Fig 8. Wear types (flank, crater) and wear–time curve

- **Flank wear** (VB) – on clearance face, by abrasion; ↑ forces, ↓ accuracy/finish
- **Crater wear** (KT, KM) – on rake face, by diffusion/adhesion at high temp.
- Other: nose wear, notch (DOC-line) wear, chipping, edge fracture, plastic deformation
- **Mechanisms**: abrasion, adhesion (attrition), diffusion, oxidation, fatigue, thermal cracking
- Stages: **I** initial rapid (edge rounding) · **II** steady (long) · **III** rapid failure

Tool life criteria (T = time of cutting until failure)

- **Flank wear VB = 0.3 mm** (uniform) or 0.6 mm (non-uniform) – ISO 3685; HSS ~0.75–1.5 rough
- Crater depth KT = 0.06 + 0.3 f (mm) · complete failure · surface finish deterioration
- ↑ cutting force / power / temperature · dimensional change · chatter or sound

Taylor's tool life equation

$$V \cdot T^{n} = C$$

- V = cutting speed (m/min), T = tool life (min), n = tool-life exponent, C = constant (speed for $T = 1$ min)
- n depends on tool material (see table); $\log V = \log C - n \log T$ → **straight line**

Tool material	n
HSS	0.08 – 0.20 (≈ 0.125)
Cemented carbide	0.20 – 0.40 (≈ 0.25)
Ceramic	0.40 – 0.70 (≈ 0.5)

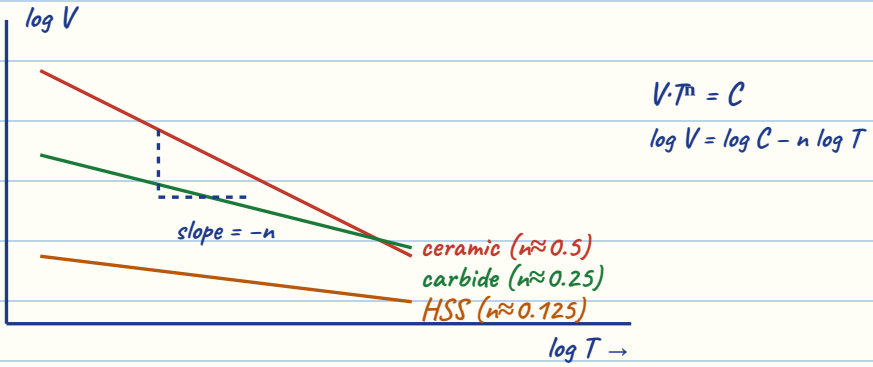


Fig 9. Tool life curves on log–log scale

Extended Taylor equation & factors affecting tool life

$$V \cdot T^{n} \cdot f^{a} \cdot d^{b} = C \quad (a \approx 0.15\text{–}0.4,\ b \approx 0.1\text{–}0.2,\ n \approx 0.1\text{–}0.5)$$

- V has the **greatest** effect, then f, then d ($V \gg f > d$)
- Also: tool material/geometry, work hardness & microstructure, coolant, rigidity of machine, BUE
- Taylor ratio: $(V_1/V_2) = (T_2/T_1)^{n}$; 20 % ↑ in V roughly halves life of HSS tool

Solved Example 2 (Taylor)

Tool life 60 min at $V = 30$ m/min and 20 min at $V = 45$ m/min. Find n, C and life at $V = 36$ m/min.

1. $V_1 T_1^{n} = V_2 T_2^{n} \Rightarrow (45/30) = (60/20)^{n} \Rightarrow n = \ln 1.5 / \ln 3 =$ **0.369**
2. $C = 30 \times 60^{0.369} = 30 \times 4.53 =$ **136**
3. $T = (C/V)^{1/n} = (136/36)^{2.71} = (3.78)^{2.71} \approx$ **36.6 min**

10. Cutting Fluids (Coolants)

Functions

- **Cool** tool & work (↓ temp) · **Lubricate** chip–tool interface (↓ friction, ↓ BUE)
- **Wash away chips** (swarf) · **protect** work/machine from rust
- ↑ tool life, ↑ surface finish, ↑ allowable V, ↓ power, ↑ dimensional accuracy

Types

Type	Composition	Main use
Straight (neat) oils	Mineral/fatty oils + EP additives (S, Cl, P)	Lubrication: low V, tough operations – threading, broaching
Soluble oils (emulsions)	Oil droplets in water 1:20–1:50 + emulsifier	Cooling + some lubrication: general turning, milling; cheap
Semi-synthetic	Small oil + chemicals in water	Mixed duty, grinding
Synthetic	Chemical (no oil) in water	High speed, grinding, best cooling, clean
Gases / solids	Compressed air, CO_2 ; graphite, MoS_2	Dry/near-dry, cast iron; high-temp lubrication

- Additives: **EP** (extreme-pressure) – sulphur, chlorine, phosphorus; anti-foam, biocide, corrosion inhibitor

Selection & application

- High V (low load) → need **cooling** → water-based; low V (high load) → **lubrication** → oil
- Steel: soluble / sulphurised oil · Al: kerosene/soluble oil · CI, brass: dry or air jet
- Mg: **never water-based (fire)** · carbide tools: dry or copious flood (avoid thermal shock)
- Methods: **flood**, mist/spray, manual, through-tool (high-pressure), minimum quantity lubrication (MQL)
- Disadvantages: cost, disposal, health (dermatitis, mist), bacteria – hence dry/MQL machining

II. Machinability

Machinability = ease with which a material can be machined, judged by **tool life, cutting force/power, surface finish, chip form.**

- Factors: work – hardness, strength, ductility, microstructure, thermal conductivity, work-hardening, composition (S, Pb, P improve; Cr, Ni, Mo reduce)
- Process – speed, feed, tool material/geometry, rigidity, cutting fluid

$$\text{Machinability index (\%)} = V_{60} \text{ of material} / V_{60} \text{ of free-cutting steel (B1112)} \times 100$$

- V_{60} = cutting speed giving 60 min tool life; free-cutting steel B1112 = **100 %**
- Typical: Al alloys 300–1500 · free-cutting brass ~ 200 · mild steel ~ 55–70 · grey CI ~ 60–80 · stainless steel ~ 25–45 · Ti alloys ~ 20

- Improve: free-machining additives (S, Pb – **resulphurised steels**), proper heat treatment (annealing), cold drawing, right tool & coolant

12. Economics of Machining

- Aim: choose **V** (and f, d) for **min. cost per piece** or **max. production rate**
- Low V → big machining time; high V → rapid tool wear → many tool changes ⇒ **an optimum V**

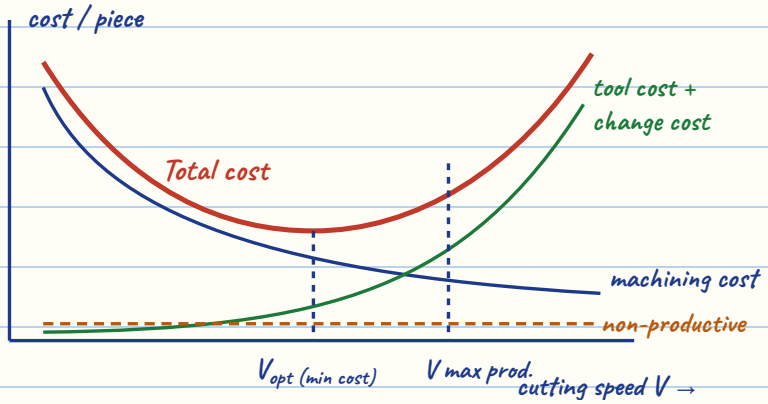


Fig 10. Cost components vs cutting speed

Cost per piece

$$C_p = C_o \cdot t_l + C_o \cdot t_m + (C_o \cdot t_c + C_t) \cdot (t_m / T)$$

- C_o = machine + operator rate (₹/min); t_l = loading/unloading time; t_m = machining time/piece
- t_c = tool change time (min); C_t = tool cost per edge (incl. regrind); t_m/T = tool changes per piece
- $T_m = \pi D L/(1000 \cdot f \cdot V)$ and $V = C/T^n$, so $t_m \propto 1/V$

Optimum tool life & speed

$$T_{min\ cost} = (1/n - 1)(t_c + C_t/C_o)$$

$$T_{max\ production} = (1/n - 1) \cdot t_c$$

- Then $V_{opt} = C / T_{opt}^{\,n}$; production time/piece = $t_l + t_m + t_c \cdot t_m/T$
- **V(max production) > V(min cost)** (since $C_t/C_o > 0$); between = **high-efficiency range**
- Min-cost used normally; max-production in emergency/bottleneck
- Higher n (ceramic) ⇒ lower optimum T ; Cost ↑ if t_c, C_t large ⇒ run slower

13. Cutting Tool Materials

Requirements

- High **hot hardness** (red hardness) · wear/abrasion resistance · toughness & shock resistance
- Low friction/affinity to work · chemical stability · thermal conductivity & low expansion · cheap, easily shaped

Material	Composition	Hot hardness limit	Characteristics / uses
HCS (high carbon steel)	0.8–1.3 % C	≈ 250 °C	Cheap, V ≈ 10 m/min; hand tools, taps, files
HSS	18W-4Cr-1V (T1); Mo type M2; Co added (T15)	≈ 600 °C	Tough, V ≈ 30–50; drills, reamers, milling cutters, complex tools
Cast alloys (Stellite)	Co 40-50, Cr 30, W 15–20	≈ 800 °C	Cast & ground, not heat-treated; V ≈ 1.5–2 × HSS; brittle
Cemented carbide	WC grains + Co binder (sintered); + TiC, TaC for steel	≈ 1000 °C	V ≈ 100–200; hard, brittle; indexable inserts; K (CI), P (steel), M grades
Coated carbide	TiN, TiC, Al_2O_3 layers (CVD/PVD) 3–10 µm	> 1000 °C	↑ wear & crater resistance, ↓ friction; 2–3 × life
Ceramics	Al_2O_3 (+ ZrO_2/TiC), Si_3N_4	≈ 1200 °C	Very hard, chemically inert, V ≈ 200–600; brittle – CI, hardened steel, rigid set-up
Cermets	TiC/TiN + Ni–Mo binder	≈ 1100 °C	Between carbide & ceramic; finishing steels, good finish
CBN	Cubic boron nitride, 2nd hardest	≈ 1400 °C	Hardened steels (> 50 HRC), superalloys; no reaction with Fe
Diamond (PCD/single)	Polycrystalline diamond compact	≈ 700 °C (in air)	Hardest; Al, Cu, plastics, composites; **not for ferrous** (carbon diffuses)

- Trend of hardness ↑ : HCS < HSS < Cast alloy < Carbide < Ceramic < CBN < Diamond (toughness ↓)
- Mnemonic: **"Hot Hardness Climbs: C-S-C-C-C-C-D"**

14. Quick Revision & Likely Exam Questions

★ Important

- $r = t/tc = \sin\varphi/\cos(\varphi-\alpha)$; $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$
- $F = Fc\sin\alpha + Ft\cos\alpha$; $N = Fc\cos\alpha - Ft\sin\alpha$; $\mu = F/N = \tan\beta$
- $Fs = Fc\cos\varphi - Ft\sin\varphi$; $Fn = Fc\sin\varphi + Ft\cos\varphi$; $Pc = Fc\cdot V$
- $\varphi = 45 + \alpha/2 - \beta/2$; $2\varphi + \beta - \alpha = C$
- $VT^n = C$; n: HSS 0.125, carbide 0.25, ceramic 0.5 ; VB = 0.3 mm
- $T(\text{min cost}) = (1/n - 1)(tc + Ct/Co)$; $T(\text{max prod.}) = (1/n - 1)\,tc$
- Heat: shear 75–80 %, friction 15–20 %, flank 2–5 %

Likely questions

1. Differentiate orthogonal & oblique cutting with neat sketches. (5M)
2. Draw a single-point tool; explain tool angles; write ASA & ORS signature. (10M)
3. Types of chips with sketches and conditions of formation. (5/10M)
4. What is BUE? Causes, effects, remedies. (5M)
5. Chip breakers – explain types with diagrams. (5M)
6. Derive relation of shear angle and chip thickness ratio. (5M)
7. Draw Merchant's circle; derive Fs, Fn, F, N in terms of Fc, Ft. (10M)
8. Merchant's two shear-angle theories; numerical on circle. (10M)
9. Heat zones and heat distribution in cutting. (5M)
10. Tool wear types, tool life criteria; Taylor's equation + numerical. (10M)
11. Cutting fluids – functions, types, selection. (5/10M)
12. Machinability & machinability index; factors. (5M)
13. Economics: derive optimum speed for min cost / max production. (10M)
14. Cutting tool materials with properties & applications. (10M)

UNIT II – Lathe, Drilling, Boring, Shaping, Slotting & Planing

23A03501 MACHINING PROCESSES · B.Tech Mech · JNTUK

I. Engine Lathe – Principle & Working

Lathe = machine tool that removes metal from a **rotating** workpiece using a **single-point** cutting tool moving linearly (parallel / perpendicular to axis).

- Work → rotates (primary cutting motion) ; tool → feed motion
- Produces **cylindrical / conical / flat / threaded** surfaces
- **Engine lathe** (centre lathe) = most common; named so as early ones were driven by steam engines
- Power from motor → belt/gears → headstock spindle → work
- Feed from spindle → gear train → **feed rod** (turning) or **lead screw** (threads)

Main parts (Fig 1)

- **Bed** – C.I. base, with **V / flat guideways**; carries all parts; rigid, damps vibration
- **Headstock** – fixed at left; has hollow **spindle**, speed gearbox, Morse taper; holds chuck
- **Tailstock** – right, slides on bed; supports long work with **dead centre**; holds drill, reamer
- **Carriage** = saddle + cross-slide + compound rest + tool post + apron
  - **Saddle** – slides on bed along Z
  - **Cross-slide** – on saddle, moves ⊥ to axis (X); gives depth of cut / facing
  - **Compound rest** – swivels on cross slide; taper turning; short feed
  - **Apron** – hangs from saddle; feed clutches, half-nut, handwheels
- **Lead screw** – ACME thread, only for thread cutting (half nut engaged)
- **Feed rod** – plain shaft with keyway; gives power feed for turning, facing

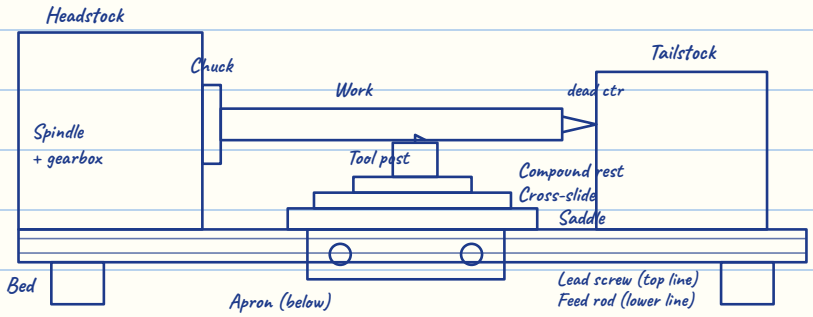


Fig 1: Engine lathe – main parts

2. Specification of a Lathe

- **Swing** over bed = max dia of work that can rotate over bed (= 2 × centre height)
- **Swing over carriage** (cross-slide) – smaller than over bed
- **Max distance between centres** = length of work between centres
- **Bed length** (overall) ; **Spindle bore** ; spindle nose type
- Range of spindle speeds (rpm) & feeds ; thread pitch range
- Motor power (kW) ; Morse taper no. of tailstock ; lead screw pitch
- Eg: **"300 mm × 1500 mm"** = swing 300, centre distance 1500

3. Types of Lathes

Type	Features / use
Speed lathe	Only headstock, tailstock, tool post on bed; 1200-3600 rpm; no feed gearbox; wood turning, spinning, polishing, centring
Engine / centre	General purpose; gear-driven; power feed + threads; medium production, jobs
Bench lathe	Small, mounted on bench; precision small parts, watches, instruments
Tool room lathe	Engine lathe + extra accuracy, wide speed range, taper attachment, collet; tool & die making
Capstan / Turret	Hexagonal turret (tailstock side) with 6 tools set in sequence; repetitive batch work. Capstan: turret on ram – light jobs, bar stock; Turret: turret on saddle – heavy, large jobs
Automatic	All tool movements by cams, no operator; single/multi spindle (screw machine, Swiss type); mass production
CNC lathe	Programmed (G & M codes), servo motors, turret of 8-12 tools; high accuracy, flexible; batch & mass

★ Important

- Capstan vs turret: capstan turret is on a **ram** (slides on saddle), turret lathe turret is on **saddle** (more rigid)
- Engine lathe = single tool at a time; turret = many tools in one setup → saves time

4. Work Holding Devices

(a) Chucks

- **3-jaw (self-centring / universal) chuck** – jaws move together by scroll; for round, hexagonal bars; quick; accuracy low
- **4-jaw (independent) chuck** – each jaw separate by screw; square, irregular, eccentric jobs; heavy duty; slow setting
- **Collet chuck** – split spring sleeve grips small bar; very accurate, bar work
- **Magnetic chuck** – holds thin ferrous parts by magnetic force, no distortion; used in grinding
- **Combination chuck** – jaws can move together or independently

(b) Others

- **Centres** – live (in headstock, rotates) & dead (tailstock; hardened, lubricated) ; 60° point; Morse taper shank
- **Face plate** – C.I. disc with slots; clamp irregular jobs by bolts, angle plate; balance with counterweight
- **Mandrel** – job with bored hole held between centres to turn outside (plain, gang, taper, expanding, stepped)
- **Lathe dog + driving plate** – drives work between centres
- **Steady rest** (fixed on bed, 3 jaws) – supports long slender job at middle / end; avoids deflection
- **Follow rest** – fixed on carriage, moves with tool; supports work against tool thrust (2 jaws); long thin shafts, threading

5. Tool Holders & Cutting Tools

- Tool held in **tool post** on compound rest (clamped by set screws; packing for height)
- Types of tool post: **single-screw (round)**, **four-bolt (square)**, **four-way turret**, **quick-change (Dickson / Multifix)**
- **Straight, left-bent, right-bent** tool holders → for roughing, left/right cuts

- **Boring bar holder** ; **Parting tool holder** ; **Threading tool holder** (spring type)
- Tool types: solid HSS; brazed carbide tip; **indexable insert** clamped on shank (clamped, not resharpened)
- Tip at **centre height** – else facing leaves a pip; threading angle wrong

6. Lathe Operations

Operation	Description
Straight turning	Reduce dia, tool ∥ axis
Facing	Flat face ⊥ axis; cross-slide feed
Taper turning	Conical surface
Thread cutting	Helical groove; tool feed = pitch/rev
Knurling	Diamond / straight pattern pressed by knurling wheels for grip (no chips)
Drilling	Drill in tailstock; work rotates
Boring	Enlarge existing hole by single point tool
Reaming	Finishing hole with reamer
Parting / cut-off	Narrow groove till job separated
Grooving / Chamfering	Form groove / bevel edge
Forming	Form tool gives contour in one plunge
Filing, polishing, spinning, eccentric turning	Minor ops

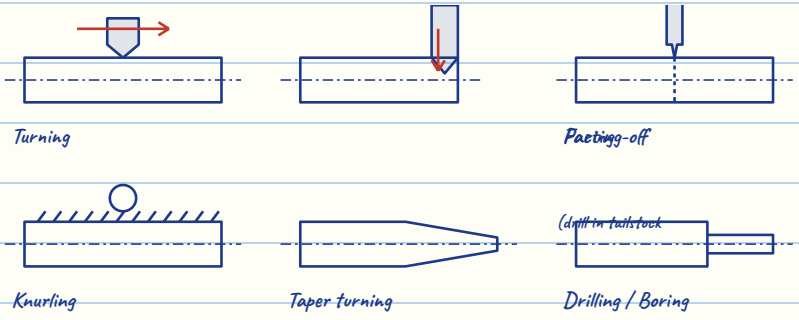


Fig 3: Basic lathe operations (schematic)

7. Taper Turning

Taper = ratio of diff in diameters to length. Conicity $K = (D - d)/l$; **half angle** α (semi-taper)

$$\tan \alpha = (D - d) / 2l$$

- D = large dia, d = small dia, l = length of tapered portion (mm)

- Taper per foot = $12(D-d)/l$; Morse taper ≈ 1 in 20 (≈ 0.05 mm/mm)

Methods

1. **Form tool** – broad nose tool ground to taper angle, plunged in; short tapers (chamfers), ≤ 10-15 mm; fast, rigid needed
2. **Compound rest swivelling** – rest set to α; hand feed by compound screw; steep tapers, short length (limited by travel); any angle
3. **Tailstock set-over** – tailstock shifted $\perp$ to axis; work axis inclined; long, gentle tapers (≤ 8-10°); centre holes get distorted; no internal tapers
4. **Taper turning attachment** – guide bar set at α on bracket behind bed; slide block moves cross-slide; accurate, long tapers (≤ 10-12°), power feed; internal too; no need to disturb tailstock
5. **Combining feeds** (CNC / hydraulic copying) – longitudinal + cross feeds simultaneously

$$\text{Tailstock set-over } S = L\,(D - d) \,/\, 2l = L \tan \alpha$$

- L = total length of work between centres, l = length of taper. If whole length tapered ($L = l$): $S = (D - d)/2$
- Alternate: $S = L \times (\text{taper ratio}) / 2$, with taper = $(D-d)/l$

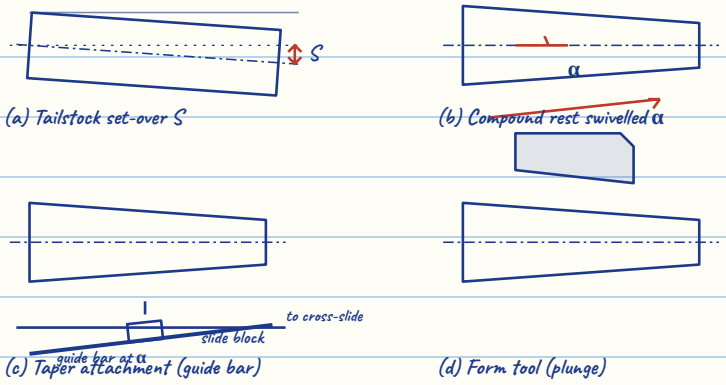


Fig 4: Taper turning methods

> ★ **Important**
>
> - Tailstock set-over: $S = L(D-d)/2l$; Attachment: best for long accurate tapers ; Compound rest: steep / short
> - Taper turn: tool tip must be at centre height

8. Thread Cutting on Lathe

- Work rotates, tool moves a distance = **pitch** per revolution → helical groove
- Tool: 60° (V, metric/Whitworth 55°) single point, ground to thread form, set **square** with a centre gauge
- Spindle → **change gears** (quadrant) → lead screw → **half nut** → carriage

Gear ratio = Driver / Driven = Pitch of job (P) / Lead of lead screw (L)

$$Z_{driver} / Z_{driven} = P / L = (N_{lead\ screw}) / (N_{spindle})$$

- Simple gearing: 1 driver (on spindle stud), 1 driven (on lead screw) + idler
- **Compound gearing** when ratio is large: (A/B) × (C/D) = P/L
- Eg: cut 3 mm pitch, lead screw 6 mm: ratio = 3/6 = 1/2 → 20T driver : 40T driven (or 30/60)
- Eg: cut 2 mm pitch, L.S. 6 mm → 1/3 → (20/40)×(30/45) ... any combination equal to 1/3
- Inch L.S. (4 TPI) to cut 10 TPI: ratio = 4/10 = 2/5 → 40T/100T (or ×20 = 40/100)

Half nut & thread chasing dial

- **Half nut** – 2 halves clamp on lead screw by lever; engaged for threads, disengaged for others
- Cut passes: tool returned after each pass (reverse spindle or withdraw), re-engage half nut for next pass
- **Thread chasing (indicator) dial** – small dial geared to lead screw on apron; shows when to engage half nut at same point so tool follows same groove
- Even no. of threads in L.S./job → engage at any line; odd → at numbered lines; fractional → specific line / keep half nut engaged
- Steps: ① set speed (slow) & gears ② set tool square, centre height ③ touch, zero cross-slide ④ light cut, check pitch with gauge ⑤ cut passes with compound rest set 29° (feeds on one flank) ⑥ finish & check by thread gauge
- Depth of V-thread ≈ 0.6495 P (ISO ≈ 0.6134 P) ; Whitworth ≈ 0.64 P

Lead = distance moved by nut in 1 rev (= pitch × starts). For multi-start thread, lead = n × pitch.

9. Lathe Attachments

- **Taper turning attachment** – long accurate tapers
- **Steady & follow rests** – support slender work
- **Milling attachment** – on cross slide, milling keyways
- **Grinding attachment** – tool-post grinder, finishes hard work
- **Copying (tracer) attachment** – hydraulic, copies template profile
- **Gear cutting attachment**, **Relieving attachment** (milling cutters, taps)

- Thread chasing dial, Quick-change gear box, Boring attachment
- Turret / multi-tool post, Cam / Eccentric turning devices, Collet attachment

10. Machining Time – Lathe

$$N = 1000\,V / (\pi D) \text{ rpm} \quad (V \text{ m/min}, D \text{ mm})$$

Turning

$$T = L / (f N) \text{ min}, \text{ with } i \text{ passes: } T = i \cdot L / (f N)$$

- L = length of cut + approach + over-travel (mm) ; f = feed (mm/rev) ; N rpm
- Number of passes i = total depth / depth per cut
- Feed per min = $f N$; MRR = $\pi D_{avg} f d V$ (mm^3/min)

Facing

$$T = (D/2) / (f N) = D / (2 f N)$$

- For constant V, N increases as D falls (CSS in CNC)

Drilling on lathe / drill press

$$T = (L + A) / (f N) \; ; \; A = 0.3\,d \text{ (approach) } [118^\circ \text{ drill: } A = 0.5\,d \cot 59^\circ \approx 0.3\,d]$$

Thread cutting & parting

$$T = i \cdot L / (P N) \text{ (feed = pitch } P \text{ per rev)}$$

Ex 1 (turning). Turn 50 mm dia, 200 mm long, V = 30 m/min, f = 0.25 mm/rev, one pass.
$N = 1000 \times 30/(\pi \times 50) =$ **191 rpm** ; $T = 200 / (0.25 \times 191) =$ **4.19 min**

Ex 2 (facing). Face 100 mm dia bar, N = 200 rpm, f = 0.2 mm/rev.
$T = 100 / (2 \times 0.2 \times 200) =$ **1.25 min**

Ex 3 (taper). Taper job D = 60, d = 40, l = 200 = full length (L = 200).
$\tan\alpha = (60-40)/(2 \times 200) = 0.05 \rightarrow \alpha =$ **2.86°** ; set-over $S =$ $200 \times 20/(2 \times 200) =$ **10 mm**

Ex 4 (threading). 2 mm pitch, 40 mm long, N = 60 rpm, 8 passes.
$T = 8 \times 40/(2 \times 60) =$ **2.67 min**

11. Drilling Machines

Drilling = making a round hole with a rotating **twist drill** (rotation + axial feed). Drill rotates, work fixed. Also gives centre drilling, reaming etc. by changing tool.

Types

- **Portable / hand drill** – electric, for site work
- **Sensitive (bench) drilling m/c** – small, up to 12–15 mm; hand feed gives "feel"; high speed
- **Upright (pillar) drill** – round column; table adjustable; medium jobs, power feed; 50 mm dia
- **Radial drilling m/c** – radial arm swings about column & moves up/down; drill head slides on arm; large & heavy jobs, hole position without moving job
- **Gang drilling m/c** – 2–6 spindles on one long table, each with different tool; sequence ops (drill, ream, tap) ; batch
- **Multi-spindle m/c** – many spindles with one head, drill many holes at once at fixed positions; mass production (auto parts)
- **Deep-hole drilling m/c** – L/D > 10 ; gun drill / BTA; high-pressure coolant through drill; horizontal; barrels, crankshafts
- **Turret / automatic / CNC drilling** m/c

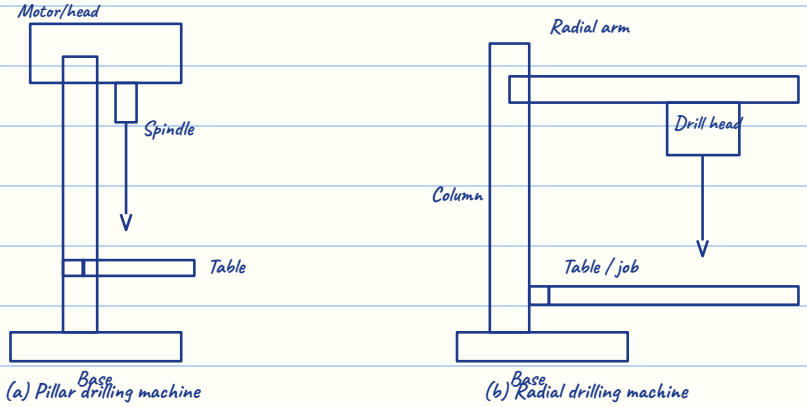


Fig 6: Pillar and radial drilling machines

Specification

- Max drill dia in steel ; table size ; spindle travel ; no. of speeds/feeds ; Morse taper no. ; swing (= 2 × distance column to spindle) ; power; radial: arm length, column dia

Twist drill nomenclature (Fig 7)

- **Point angle** (118° general; 135° hard/steel) ; **lip clearance** 8-12° ; **helix angle** 24-32° (rake)
- **Chisel edge angle** 115-135° ; **web** (thickens to shank) ; **flutes** (chip removal, coolant)

- **Land**, **margin** (guides), **body clearance**, **lips** (cutting edges), **shank** (straight / Morse taper), **tang** (drives drill)

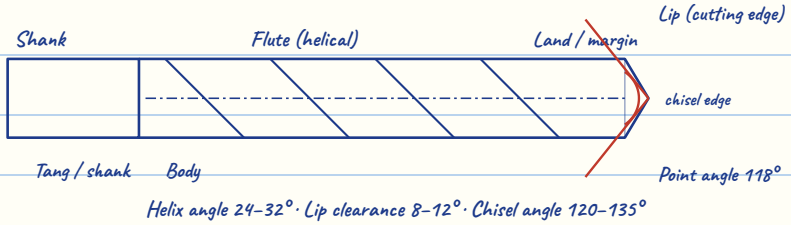


Fig 7: Twist drill – nomenclature

Tool & work holding (drilling)

- **Drill chuck** (jaws, key) for straight shank ; **sleeve / socket** & **drift** for Morse taper shank
- Work: **machine vice, V-block, T-bolts & clamps, angle plate, jig**

Operations on drilling m/c (Fig 8)

Operation	Meaning
Drilling	Make new hole
Reaming	Finish hole to accurate size, good finish (multi-tooth reamer, small allowance 0.2–0.5 mm)
Boring	Enlarge / correct hole by single-point bar
Counter-boring	Enlarge end of hole, square shoulder (for bolt heads)
Counter-sinking	Conical enlargement for flat-head screw (60°/90°)
Spot-facing	Smooth flat seat around hole for nut/washer
Tapping	Cut internal threads with tap
Trepanning / Centring	Cut annular groove for big holes ; centre drill

Drilling time

$$N = 1000\,V / (\pi\, d)\ ;\ T = (L + A) / (f\, N)\ ;\ A \approx 0.3\, d$$

- L = depth of hole (for blind); through hole: L = thickness + approach + breakthrough (≈ 1–2 mm)
- $MRR = (\pi\, d^2/4)\, f\, N$

Ex 5. Drill 20 mm hole, 40 mm deep, V = 25 m/min, f = 0.2 mm/rev.
$N = 1000\times25/(\pi\times20) = 398$ rpm ; $A = 0.3\times20 = 6$ mm ; $T = (40+6)/(0.2\times398) =$ **0.58 min**

12. Boring Machines

Boring = enlarging & finishing an existing hole with **single point** tool on a boring bar (concentric, straight, accurate).

- **Horizontal boring m/c (HBM)** – horizontal spindle; table (X, Z), head moves on column (Y); outboard support; big castings, gear boxes, engine blocks; also drilling, milling, facing
  - Types: table, planer, floor, multiple-head
- **Vertical boring m/c (VBM / vertical turret lathe)** – vertical spindle, **rotating table** (like inverted lathe); heavy, large dia, short work (flywheels, rings, turbine casings); cross rail, 2 heads
- **Jig boring m/c** – very high precision (±2 μm); coordinate positioning by micro-scales; temp. controlled room; jigs, fixtures, dies
- **Precision boring** – diamond / carbide tools, high speed light cut; engine cylinders, bearings

Boring tools

- Single-point bit in **boring bar** (set screw) ; **offset boring head** (micro adjust dia) ; **line boring** (several coaxial holes) ; **back boring** ; **floating** reamer type tool
- Bar overhang short; $L/D \leq 4$ for steel bar (carbide bars up to 7)

Drilling	Boring
Makes new hole	Enlarges existing hole
Multi-edge twist drill	Single-point tool
Poor concentricity	Corrects position / alignment

13. Shaper

Shaper = reciprocating m/c; **single-point tool** moves to-and-fro (cutting on forward stroke), work fixed on table gets **intermittent feed** at the end of return stroke. Produces flat (horizontal, vertical, inclined), grooves, slots, gear teeth.

Parts

- **Base** – C.I., supports all
- **Column** – houses drive; guideways on top for ram
- **Ram** – reciprocates in dovetail guides, carries tool head
- **Tool head** – **down feed screw, swivel base (angular cuts), apron with clapper box, tool post**
- **Clapper box** – tool lifts on return stroke (no drag)
- **Cross-rail** (elevates), **saddle** (cross feed), **table** (T-slots, work vice)

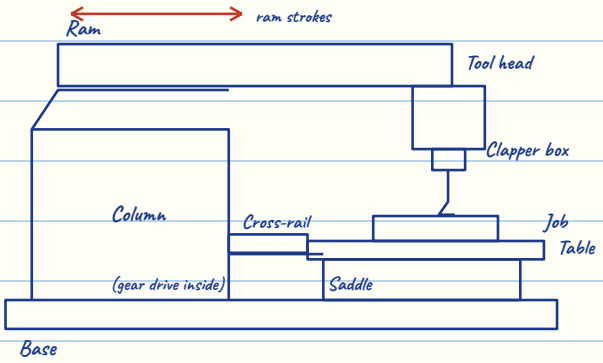


Fig 9: Shaper – schematic with main parts

Quick-return mechanisms (Fig 10)

- (a) **Crank & slotted link** (most common): bull gear + sliding block (crank pin) → slotted link (rocker arm) pivoted at bottom → ram via link
- (b) **Whitworth QR** – crank centre inside slotted lever pivot circle; used in heavy shapers/slotters; ratio fixed by geometry
- (c) **Hydraulic shaper** – oil piston, constant cutting speed, smooth, stepless speed, overload safe

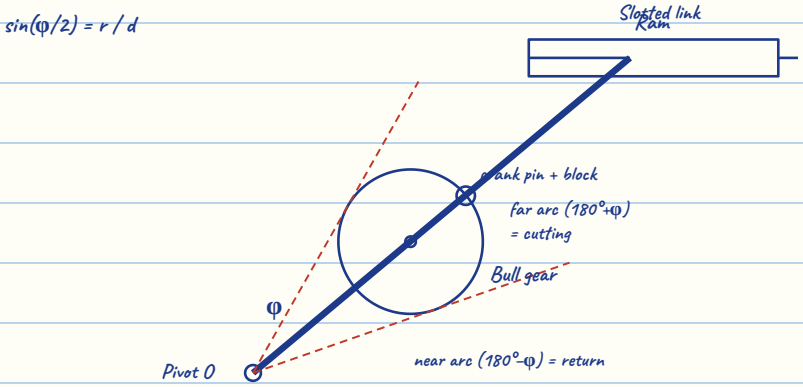


Fig 10: Crank & slotted-link quick-return mechanism (O to bull-gear centre = d, crank radius = r)

- Cutting stroke: crank turns through angle $(180° + \varphi)$; return: $(180° - \varphi)$ → return faster at const. bull-gear speed
- Return ratio: $\sin(\varphi/2)$ = crank radius / distance between centres

$$\text{Quick return ratio} = \text{cutting time / return time} = (180 + \varphi) / (180 - \varphi) \quad (> 1, \approx 3{:}2)$$

Specification & types

- Max length of stroke (150–900 mm) ; table size ; max vertical travel ; power ; no. of strokes/min
- **Types:** crank / geared / hydraulic (drive) ; horizontal, vertical (slotter), travelling head (long jobs) ; push-cut / draw-cut ; standard / universal (table tilts)

Work holding & tools

- Machine vice, T-bolts & clamps, V-blocks, angle plate, special fixtures, stop pins
- Tools: round-nose roughing, finishing broad tool, side-recess, parting, form tool ; HSS ; relief for return stroke

Operations

- Horizontal, vertical, angular surfaces ; **slots, grooves, keyways, T-slots, dovetails** ; irregular / contour ; gear teeth (rack) ; splines

Machining time

- Length of stroke L = job length + 20–30 mm clearance (approach + overtravel)
- Let m = return time / cutting time = 1/(ratio) , V = cutting speed (m/min)

$$\text{Strokes per min } N = 1000\,V / [\,L\,(1+m)\,] \quad (L \text{ in mm})$$

$$T = W / (f \times N) \text{ min} \quad (W = \text{width} + \text{clearance},\ f = \text{feed mm/stroke})$$

- Number of cuts i = total depth / depth per cut → multiply
- Or time/stroke = $L/V_c + L/V_r$

Ex 6. Shape plate 300 mm wide, 180 mm long ; clearance 20 → L = 200 mm. V = 15 m/min, ratio 3 : 2 (m = 2/3), f = 1.5 mm/stroke.
$N = 1000\times15/[200\times(1+0.667)]$ = **45 strokes/min** ; $T = 300 / (1.5\times45)$ = **4.44 min**

★ *Important*

- Cutting on forward stroke only ; feed at end of return stroke ; ram stroke adjusted by crank-pin radius (position by ram screw)
- Ram speed variable along stroke (max at mid) ; ratio of return ≈ 2:3

14. Slotter (Slotting Machine)

Slotter = vertical shaper; ram moves **vertically**, cutting on **downward** stroke; tool is single-point; work on rotary table with **3 feeds**: longitudinal, cross, rotary.

- **Parts:** base, column (hollow, houses drive), saddle, cross-slide, **circular (rotary) table**, ram + tool head, drive (Whitworth QR or hydraulic)
- Ram can be tilted slightly ($\leq 10°$) for tapered cuts
- **Types:** puncher (heavy), precision (tool-room), tool-room slotter
- **Operations:** internal keyways, splines, slots, internal gears, circular & contour surfaces, square / hexagonal holes

- Whitworth mechanism: crank pin rotates; slotted-link end (pivot at centre of gear) ; ram moves down slow (cut), up fast

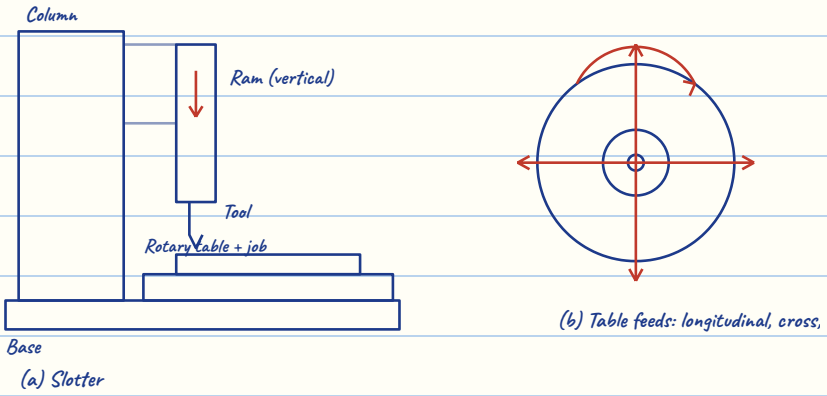


Fig 11: Slotting machine and its three table feeds (top view of table)

15. Planer (Planing Machine)

Planer = for large, heavy flat surfaces. **Work reciprocates** on table; tool fixed (feeds intermittently across at the start of the return stroke). Cutting on forward stroke, tool lifts on return (clapper box).

Parts

- **Bed** (long, with V & flat ways) ; **Table** (T-slots, holds work, moves on bed)
- **Housings / columns** (one or two, vertical) ; **Cross-rail** (elevating, carries tool heads)
- **Tool heads** – 2 on rail, 1–2 on housings (side heads) ; each with clapper box, tool post, swivel
- **Driving mechanisms** (drive) ; **Feed** mechanism ; reversing dogs

Drives of planer

- **Open and cross belt drive** – 2 belts, forward (open, slow) & return (cross, fast); ratio 1:3–1:4 ; old
- **Reversible motor drive** – D.C. motor via gears & rack-pinion; dogs reverse the motor; Ward-Leonard control
- **Hydraulic drive** – piston, smooth, speed constant, high loads
- **Crank drive** – small planers
- Table speed: cutting 6–30 m/min ; return 2–4 × cutting

Types

- **Double-housing** – two housings joined by cross-rail ; rigid ; width of work limited by housing gap ; most common
- **Open-side (single-housing)** – one housing, overhanging cross-rail ; very wide work ; less rigid

- **Pit planer** – tool bridge moves over fixed table in pit ; very large & heavy jobs
- **Edge (plate) planer** – plate edges for welding preparation (bevel) ; work clamped & tool moves along edge
- **Divided-table planer** – 2 tables, setting up one while other cuts

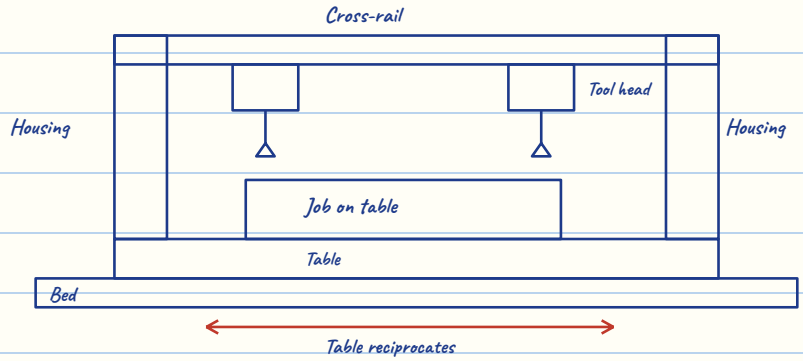


Fig 12: Double-housing planer

Specification

- Max length of table travel ; max width between housings ; max height under rail ; table size ; cutting & return speeds ; no. of tool heads ; motor kW

Tools, work holding, operations

- Tools: like shaper but heavier, with larger cross-section ; carbide-tipped, broad finishing tool
- Holding: T-bolts, clamps, **planer centres**, stops, angle plates, toe-dogs, jacks
- Ops: flat, horizontal / vertical / inclined surfaces, slots, T-slots, dovetails, grooves – machine beds, tables, large castings, ways of lathes ; several jobs simultaneously

Machining time (planer)

$$T = (W / f) \times [\, L / V_c + L / V_r \,]$$ (L = table travel (job + overtravel), V in m/min)

Ex 7. Planer job 1000 mm long, 600 mm wide ; overtravel 100 mm → $L = 1.1$ m ; $V_c = 12$ m/min, $V_r = 36$ m/min, $f = 2$ mm/stroke.

Time/double stroke = $1.1/12 + 1.1/36 = 0.0917 + 0.0306 = 0.1222$ min ; strokes = $600/2 = 300$

$T = 300 \times 0.1222 =$ **36.7 min**

16. Comparison: Shaper vs Planer vs Slotter

Point	Shaper	Planer	Slotter
Moving part	Tool (ram)	Work (table)	Tool (vertical ram)
Job size	Small / medium	Large, heavy	Medium (internal)
Cutting stroke	Forward, horizontal	Table forward, horiz.	Downward, vertical
Feed	Table, at end of return	Tool, at start of return	Longit., cross, rotary
Tool heads	1	2–4	1
Drive	Crank-slotted link / hydraulic	Reversing motor / belt / hydraulic	Whitworth / hydraulic
Rigidity, cut	Light cuts	Heavy, accurate	Light / medium
Use	Flat, slots, keyways	Beds, tables, ways	Internal keyways, gears
Cost, floor	Low	High, large floor	Medium

★ **Important**

- Shaper: tool moves, work feeds ; Planer: work moves, tool feeds
- Mnemonic: **S**haper = **S**mall, **P**laner = **P**onderous, **S**lotter = **S**tanding (vertical)
- Quick return ratio for shaper ≈ 3:2 (cutting : return)

17. Quick Revision & Likely Exam Questions

Formulas

$N = 1000V/\pi D$; $T(\text{turn}) = L/fN$; $T(\text{face}) = D/2fN$; $T(\text{drill}) = (L+0.3d)/fN$; $\tan\alpha = (D-d)/2l$; $QRR = (180+\varphi)/(180-\varphi)$

Likely questions

1. Sketch an engine lathe & explain its main parts. (long)
2. Specify & classify lathes ; lathe operations & attachments.
3. Describe work holding devices (chucks, centres, mandrels, rests) used on lathe.
4. Explain taper turning methods ; derive tailstock set-over.
5. Explain thread cutting on lathe – change gears, half nut, chasing dial.
6. Derive machining time for turning, facing & drilling; solve numerical.
7. Classify drilling machines ; radial drill sketch ; twist drill nomenclature.
8. Horizontal, vertical & jig boring m/c ; boring tools.
9. Shaper : parts, quick-return mechanism, machining time.
10. Slotter vs shaper ; planer types & drives.

UNIT III – Milling, Grinding, Finishing, Jigs & Fixtures

23A03501 Machining Processes · B.Tech Mech · Handwritten exam notes

1. MILLING MACHINE

Milling = machining with a **multi-point rotating cutter** fed against a work moving (table) – each tooth removes a small chip intermittently.

Principle of working

- Cutter rotates on spindle/arbor (**main motion**)
- Work clamped on table → **feed** (long., cross, vertical)
- Cutting speed $V = \pi DN/1000$ m/min ; feed → per tooth f_t, per rev, per min

Specifications of a milling m/c

- Table size (length × width) & max. travels – long., cross, vertical
- Spindle taper (ISO 40/50), spindle speeds range & no. of feeds

Parts of column & knee type

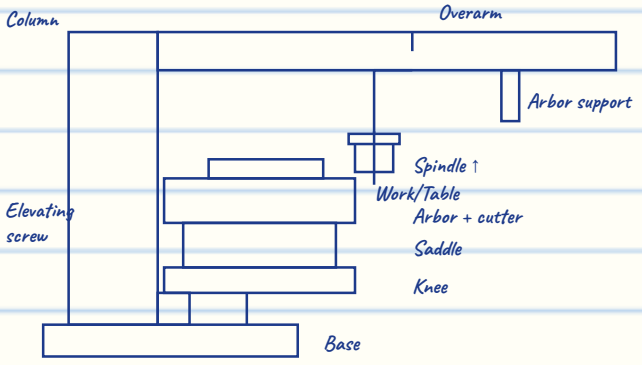


Fig 1: Column & knee milling machine (schematic)

- **Base** – foundation, holds coolant; **Column** – houses spindle drive, speed gear box
- **Knee** – slides on column, vertical motion by elevating screw
- **Saddle** – on knee, cross-feed; **Table** – on saddle, long. feed (T-slots)
- **Arbor** – shaft holding cutter; **Arbor support** – avoids deflection

2. Classification of milling machines

Type	Features / use
Column & knee	Most common, tool-room & general work. (a) **Plain** (horizontal spindle, 3 motions) (b) **Universal** (table swivels ±45° → helical work) (c) **Vertical** (spindle vertical, end mills, T-slots, dies)
Bed type	Table on rigid bed, only long. motion; spindle head moves vertically; heavy, mass production. Simplex / duplex / triplex
Planer type	Like planer; 2–4 heads on cross-rail & columns; heavy big jobs (machine beds)
Special type	Rotary table, drum, profile (copy), pantograph, gear hobbing, thread, tracer, CNC milling

3. Milling methods

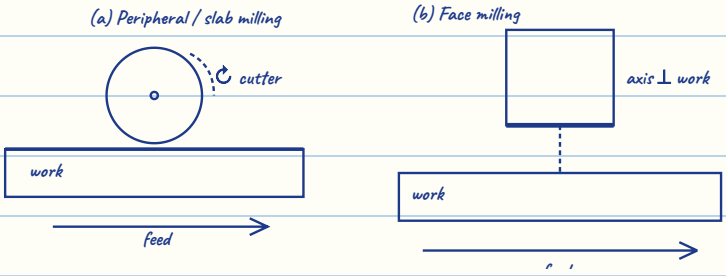


Fig 2: Peripheral and face milling

- **Peripheral (slab) milling** – teeth on periphery; cutter axis ∥ work surface; plain/helical cutter, horizontal mill; good finish for flat/wide surface
- **Face milling** – teeth on face & periphery; axis ⊥ to work; large flat surfaces, high MRR, inserted-tooth cutters
- **End milling** – slots, pockets, profiles (vertical mill)

Up (conventional) vs Down (climb) milling

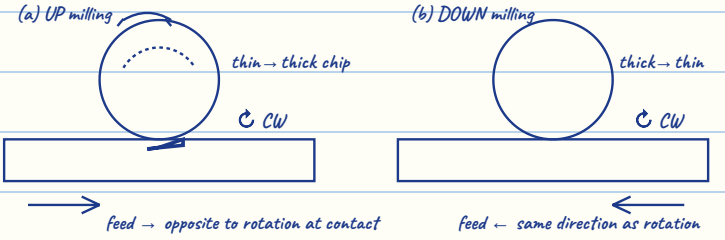


Fig 3: Up vs down milling (cutter ↻, work direction differs)

Point	Up (conventional)	Down (climb)
Feed vs rotation	Opposite at contact	Same direction
Chip thickness	Zero → max (thin start)	Max → zero
Cutting force	Lifts work, tends to loosen	Pushes work down on table
Surface finish	Poorer (rubbing, BUE)	Better
Tool life	Less (rubbing, scale)	More
Backlash	Not harmful	Needs **backlash eliminator**
Use	Rough, castings, forgings (hard skin)	Finishing, thin & rigid, CNC

4. Milling cutters

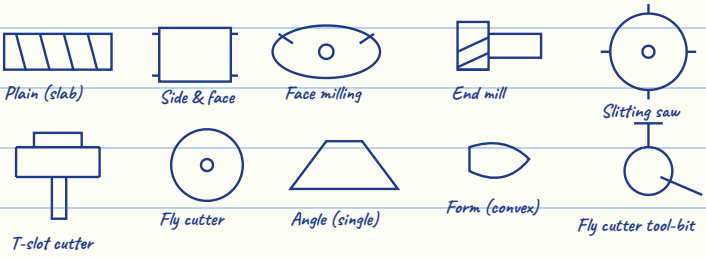


Fig 4: Common milling cutters

- **Plain (slab)** – cylindrical, teeth on periphery; helical teeth → smooth cut
- **Side & face** – teeth on periphery + one/both sides; slots, steps; **staggered** type for deep slots
- **Face mill** – large dia, inserted carbide tips; **End mill** – shank, teeth on end+periphery
- **Slitting saw** – thin (0.5–5 mm), parting & deep narrow slots
- **T-slot cutter** – end mill + small neck & wide head; cuts T-slots after slot
- **Angle** – single/double angle (45°,60°) → dovetails, V-grooves, ratchet
- **Form cutters** – convex, concave, corner-rounding, gear (involute) cutters
- **Fly cutter** – single point bit in arbor; cheap special profile / facing

Cutter nomenclature (plain milling cutter)

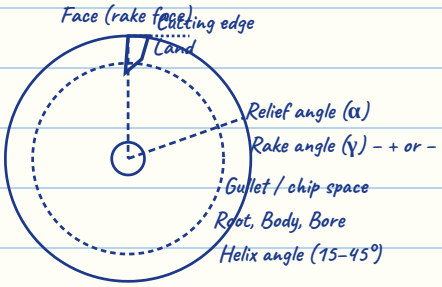


Fig 5: Nomenclature of milling cutter

5. Milling operations

- **Plain/slab milling** – flat surface · **Face milling** – flat ⊥ axis
- **Side milling** – vertical side face · **Straddle milling** – two parallel sides at once (2 side cutters)
- **Gang milling** – several cutters on one arbor, complex profile in one pass
- **Angular milling** – V-grooves, dovetail · **Form milling** – irregular contour, gears

6. Accessories & attachments

- **Arbor** – stub/long; Morse or ISO taper shank; key, spacers, bearing bush, draw-bar
- **Vices** – plain, swivel, universal (tilt) – hold work on table
- **Dividing (indexing) head** – divides periphery of work in equal parts; spiral-milling; plain, universal, optical
- **Rotary table** – circular slots, arcs, circular profile; worm & wheel drive (90:1)
- **Vertical milling head** – converts horizontal → vertical; swivels at angle
- **Slotting attachment** – converts rotary → reciprocating; keyways, splines, internal slots

7. Indexing (dividing head)

Indexing = dividing work periphery into **N equal parts** (gear teeth, splines, bolt heads, flutes) by turning work through equal angles 360°/N.

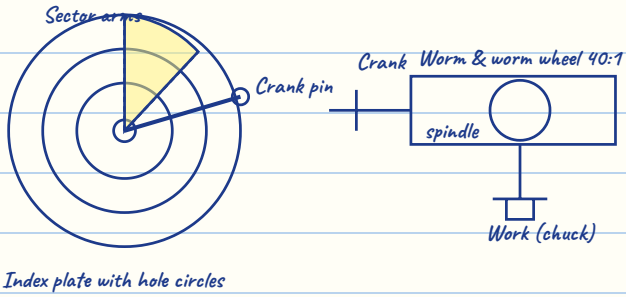


Fig 6: Dividing head – index plate & sector arms

(a) Direct indexing

- Index plate fixed on spindle (24 holes); plunger pin; worm disengaged
- Quick; only divisions = factors of 24 (2, 3, 4, 6, 8, 12, 24)

(b) Simple (plain) indexing

- Worm & wheel ratio **40:1** → 40 turns of crank = 1 rev of work
- Crank turns for N divisions:

$$n = 40 / N \text{ turns of crank}$$

- Index plates (B&S): P1: 15,16,17,18,19,20 · P2: 21,23,27,29,31,33 · P3: 37,39,41,43,47,49
- Method: fraction → full turns + (holes ÷ circle); set **sector arms** to count holes

Ex 1: N = 7 → $40/7 = 5\,{}^{5}/_{7} = 5 + {}^{15}/_{21}$ → **5 full turns + 15 holes in 21-hole circle**

Ex 2: N = 9 → $40/9 = 4\,{}^{4}/_{9} = 4 + {}^{12}/_{27}$ → **4 turns + 12 holes in 27-circle**

Ex 3: N = 28 → $40/28 = 10/7 = 1\,{}^{3}/_{7} = 1 + {}^{9}/_{21}$ → **1 turn + 9 holes in 21-circle**

Ex 4: N = 24 → $40/24 = 1\,{}^{2}/_{3} = 1 + {}^{10}/_{15}$ → 1 turn + 10 holes in 15-circle

(c) Compound indexing

- Two movements: crank rotated in plate (as simple) + **plate itself rotated** by 2nd pin (plate unlocked)
- $40/N = (a/b) \pm (c/d)$ → gets divisions not possible by simple e.g. 77, 91, 99

(d) Differential indexing

- Index plate is **geared to spindle** (gear train) – plate rotates automatically while crank turns
- Choose approx. number A (easy by simple indexing, near N):
- Crank turns = 40/A ; **Gear ratio = $40(A - N)/A$ = Driver/Driven**
- A > N → plate rotates same as crank; A < N → opposite (idler added)

(e) Angular indexing

- Dividing work by angle θ°: 1 turn of crank = 360/40 = **9°**

$$n = \theta^\circ / 9 \text{ (also 1 hole in 18-circle = } 0.5^\circ)$$

Ex: index 45° → 45/9 = 5 full turns; for 17°30′ = 17.5/9 = $1\,{}^{17}/_{18}$ → 1 turn + 17 holes in 18-circle (using 1 turn=9°; 1 hole in 18-circle = 0.5°)

Milling time & formulae

$$V = \pi DN/1000 \; ; \; f_m = f_t \times z \times N \text{ (mm/min)}$$

$$T = (L + A + O) / f_m \text{ min}$$

- L = length of work; A = approach; O = overrun (≈ 2–5 mm)
- Slab milling: $A = \sqrt{[d(D - d)]}$ (d = depth of cut, D = cutter dia)

- Face milling: $A = O = D/2$ (central) or $\approx \frac{1}{2}[D - \sqrt{(D^2 - W^2)}]$ (W = width)
- **MRR = $w \times d \times f_m$** (mm^3/min); N = rpm, z = teeth

> ★ *Important*
> - Simple indexing: $n = 40/N$ – learn 7, 9, 28 examples
> - Up milling vs down milling table – 5 marks every year
> - Time = $(L + A + O)/(f_t z N)$

GRINDING

8. Grinding machine – theory

Grinding = material removal by **abrasive grains** bonded in a wheel rotating at high speed (30–45 m/s); **very small chips**, fine finish (Ra 0.2–1.6 µm), tolerance ±0.002 mm; machines hard/hardened materials.

- Each abrasive grain = **negative-rake** (−30° to −60°) random cutting edge
- Chip formation: **rubbing → ploughing → cutting** (grain first rubs, then ploughs plastically, then cuts a chip)
- **Wheel wear**: (i) attritious wear – grain flattening (ii) grain fracture (iii) bond fracture / grain pull-out
- **Self-sharpening**: dull grain → force ↑ → grain fractures (new edges) or breaks out of bond → fresh grain exposed. Needs right **grade**

Classification of grinding machines

- **Precision**: cylindrical (external), internal, surface, centreless, tool & cutter, special (thread, gear, form, crankshaft, cam)

9. Cylindrical grinding machine

- Work between centres/chuck, rotates; wheel rotates (opposite at contact) ; table reciprocates
- **Plain** – wheel head fixed, table reciprocates (or traverse) ; **Universal** – table & wheel head swivel, internal grinding attachment, tapers
- Operations: **traverse** (long. feed), **plunge** (in-feed, no traverse – shoulders, forms), taper grinding

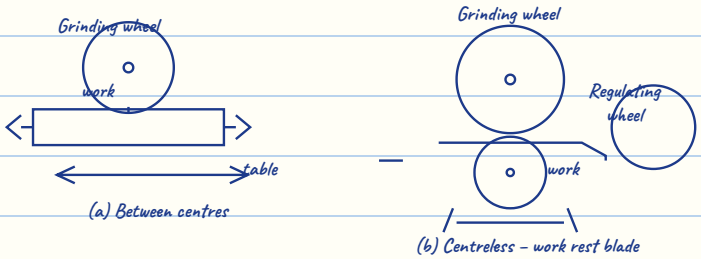


Fig 7: Cylindrical (centres) & centreless grinding

Centreless grinding

- No centres/chuck; work supported on **work-rest blade** between **grinding wheel** (fast) & **regulating wheel** (rubber-bonded, slow 20–100 rpm, tilted 1–5°)
- Regulating wheel controls work rotation & axial feed: $F = \pi D N \sin \alpha$
- + : no clamping, thin/long bars, high production, automatic, no deflection
- – : not for keyways / interrupted surfaces; setup costly

10. Surface grinding machine

- Produces flat surfaces; work on **magnetic chuck**/vice
- 4 types by wheel axis & table:

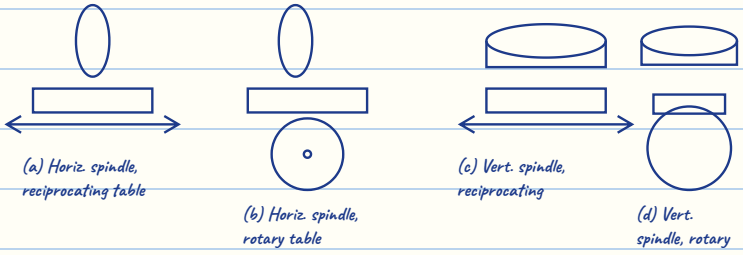


Fig 8: Four types of surface grinders

- (a) Horizontal spindle + reciprocating table – periphery of wheel; common, good finish
- (b) Horizontal spindle + rotary table – cylindrical/ring surfaces
- (c) Vertical spindle + reciprocating – cup/segmental wheel, large area, fast MRR
- (d) Vertical spindle + rotary table – production grinding of rings, washers

Internal grinding & Tool and cutter grinder

- **Internal**: small wheel in bore; chucking (work rotates), planetary (work fixed, wheel orbits), centreless internal
- **Tool & cutter grinder**: sharpens milling cutters, reamers, taps, drills, single-point tools; table swivels, work head, tailstock, universal wheel head; **cup / dish / saucer** wheels; tooth rest finger holds tooth at height of centre

11. Abrasives

Abrasive	Features / use
Aluminium oxide Al_2O_3	Artificial; tough; steels, steel alloys, high tensile; brown, white (friable), pink, ruby. Symbol A
Silicon carbide SiC	Harder but brittle; **cast iron, brass, Al, ceramics, carbide, non-ferrous**; green / black. Symbol C
CBN (cubic boron nitride)	Hardness ≈ diamond; HSS, hardened steels, Ni-superalloys; heat-resistant; costly
Diamond	Hardest; carbide, ceramics, glass, stone; not for steel (carbon affinity)
Others	Natural: emery, corundum, quartz, garnet; Zirconia alumina

➤ Mnemonic: **Steel → A**, **Cast iron/Carbide → C**

12. Bonds

Bond (symbol)	Features / uses
Vitrified (V)	Clay/glass fused; porous, strong, rigid, not affected by water/oil; 75 % wheels; max 35 m/s. Fragile (shock)
Silicate (S)	Sodium silicate; mild, grain released easily; cool cutting – edge tools, large wheels
Shellac (E)	Elastic; good finish; cam, crankshaft, roll grinding, thin wheels
Rubber (R)	Vulcanized rubber; elastic, thin cut-off wheels, regulating wheel of centreless
Resinoid (B)	Bakelite/synthetic resin; strong, up to 45–80 m/s; snagging, cut-off, rough grinding
Metal (M)	Bronze/nickel; for diamond & CBN wheels; long life

13. Grinding wheel specification

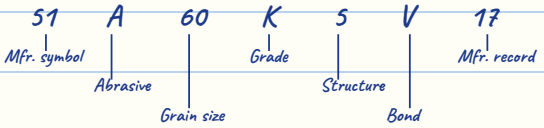


Fig 9: Marking system A 60 K 5 V (IS 551 / ISO)

➤ **A** = abrasive: A (Al_2O_3), C (SiC)

➤ **60** = grain size (mesh/in): coarse 10–24, medium 30–60, fine 70–180, very fine 220–600

➤ **K** = grade (bond strength): A–H soft, I–P medium, Q–Z hard

➤ **5** = structure (spacing): 1–8 dense (close), 9–15 open (wide)

- **V** = vitrified bond
- Read as: Al_2O_3 grain, medium-fine, medium grade, medium-dense, vitrified

★ *Important*

- Hard material → **soft grade, fine grain**; Soft/ductile material → hard grade, coarse grain, open structure
- Large contact area → soft grade, open structure; fine finish → fine grain

Wheel shapes (IS 2370)

- Straight (1), cylinder, tapered, straight cup, flaring cup, dish, saucer, segments, mounted points

Glazing & loading

- **Glazing**: grains dull, shiny surface; too hard grade → high temp, burn → dress
- **Loading**: pores clogged with metal chips (soft/ductile metals: Al, Cu); remedy – coarser/open wheel, coolant, dressing

14. Truing, dressing & balancing

Truing	Dressing
Restores **geometrical shape** & concentricity of wheel	Restores **cutting ability** – removes glazed grains / loaded chips
Done for new wheel / formed profile	Exposes fresh sharp grains
Tool: single-point **diamond**, diamond roll, form crushing roll	Tool: **star (Huntington) dresser**, abrasive stick, dressing stone, diamond
Removes more material	Removes only little layer

- **Balancing**: static balancing on balancing stand with ring-test; counter weights in flange slot → avoids vibration, chatter, wheel burst
- **Ring test** (tap – clear ring = no crack); use **blotters** & flanges (≥ ⅓ dia); wheel guard

FINISHING PROCESSES

15. Lapping

Lapping = abrasive finishing; loose abrasive **(Al_2O_3, SiC, diamond dust in oil/grease paste)** between work & a soft **lap** (cast iron, copper, lead), relative rubbing motion → removes 0.003–0.03 mm, tolerance ±0.0003 mm, Ra 0.025–0.2 μm.

16. Honing

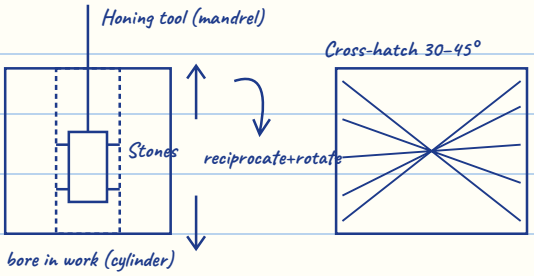


Fig 10: Honing – rotating + reciprocating hone & cross-hatch pattern

- **Honing** = low-speed finishing of **cylindrical bores** with bonded abrasive stones (hone) mounted in an expanding head; rotation (≈ 100–150 rpm) + reciprocation → **cross-hatch** 30–60° (retains oil)
- Removes 0.005–0.05 mm; Ra 0.1–0.4 µm; corrects **roundness, taper, straightness**
- Uses: engine cylinders, gun barrels, hydraulic cylinders, gears (hone)
- **Superfinishing (microfinishing)**: fine stone, light pressure (0.1–0.3 MPa), short **oscillating stroke** (1–5 mm) at 10–50 Hz + slow work rotation; lubricant (kerosene) → very smooth, Ra 0.01–0.1 µm; removes smeared layer; crankshaft, bearing races

17. Broaching

Broaching = a multi-tooth tool (**broach**) pushed/pulled past/through work in a **single stroke**; each tooth is higher than previous (**rise per tooth** 0.02–0.1 mm) → progressive cutting, finishes work in one pass.

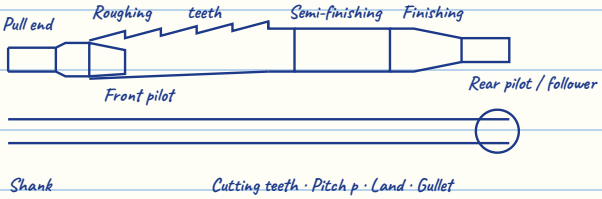


Fig 11: Parts of a broach

- **Parts**: pull end/shank, neck, front pilot, **roughing teeth**, **semi-finishing**, **finishing teeth**, rear pilot, follower rest/retriever
- **Nomenclature**: pitch ($p \approx 1.76\sqrt{L}$, L = cut length), land, gullet (chip space), face angle (rake 8–15°), back-off/clearance 1–3°, rise per tooth
- **Push type** – short, less than 8× dia (buckling) ; **Pull type** – tension, long, more accurate

Types of broaching

- **Internal**: round holes, keyways, splines, polygons (pull broach through pre-drilled hole)
- **Surface (external)**: flat, slots, contour, gun-slides; pull-up/pull-down, push; **continuous** (chain, rotary)
- **+ Advantages**: high production, high accuracy (±0.01 mm), good finish, roughing+finishing in 1 stroke, unskilled operator
- **– Limitations**: costly special tool, only through holes/surfaces, large force, uneconomical for small batches, no blind holes

18. Comparison: Grinding vs Lapping vs Honing

Point	Grinding	Lapping	Honing
Abrasive	Bonded wheel	**Loose** in paste on lap	Bonded stones
Speed	High (30 m/s)	Low	Low-medium
Pressure	High	Very light	Moderate
Metal removal	0.05–0.5 mm	0.003–0.03 mm	0.005–0.05 mm
Finish Ra (µm)	0.2–1.6	0.025–0.2	0.1–0.8
Surfaces	All (flat, cyl.)	Flat, cyl, spherical	Only bores (mostly)
Heat/stress	High, burn risk	Negligible	Low
Corrects	Size, finish	Flatness, finish	Roundness, taper, finish

JIGS & FIXTURES

19. Introduction & purposes

Jig = work-holding device that **locates, holds AND guides the tool** (bushes) – drilling, reaming, tapping. **Fixture** = locates & holds work, tool is **set** by feelers/set blocks, but **not guided** – milling, turning, grinding, welding.

Jig	Fixture
Guides tool (drill bush)	Only positions; tool set by gauge
Light, not clamped to table	Heavy, bolted to table (T-slot key)
Drilling, reaming, tapping	Milling, turning, grinding, broaching, welding

- **Purposes / uses**: eliminate marking-out & setting, **interchangeability**, accuracy, ↑ production, ↓ cost & time, unskilled labour, safety, less inspection, enable automation

20. Principles of location – 3-2-1 rule

- Free body has **12 degrees of freedom** (6 translational ±X,±Y,±Z + 6 rotational, i.e. 6 axial + 6 radial); clamping restrains the rest.
- **3-2-1 principle**: 3 pins on **primary** plane (restrain 3: 1 translation + 2 rotation) ; 2 pins on **secondary** plane (1 translation + 1 rotation) ; 1 pin on **tertiary** plane (1 translation) = **6 DOF**

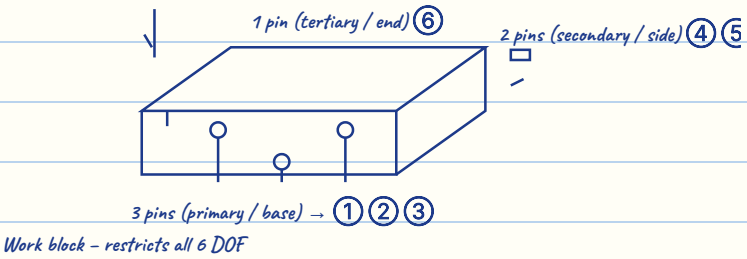


Fig 12: 3-2-1 principle of location (schematic)

Primary plane: **widest** area, 3 pins as far apart as possible → stability.
Secondary: 2 pins (long side). Tertiary: 1 pin (end).

Locating elements

- **Pins**: fixed, **button** (flat/round head), conical, **diamond pin** (relieved cylindrical, for hole positioning with 1 round + 1 diamond pin)
- **V-blocks**: centralise round work (self-centring, 90° / 120°), long/short
- **Rest buttons, pads, plates**, adjustable **jack pins** (rough castings), **nest**/cup locators
- **Locating from hole**: 1 cylindrical pin + 1 diamond pin; surface: ground flats

Principles of location & clamping

★ Important

- Locators on **machined** faces (datum), few, **sharp edges avoid**, chip-clearance, easily cleaned
- Clamp **against** locators – force directed on **supports**, not on unsupported work
- Clamp must be **quick, simple, strong**, not damage or distort work, not interfere with tool path
- Cutting force should be taken by locators/ fixed jaw – not by clamp
- Clamp: adequate force, **foolproof** (poka-yoke), loading/unloading easy

21. Clamping devices

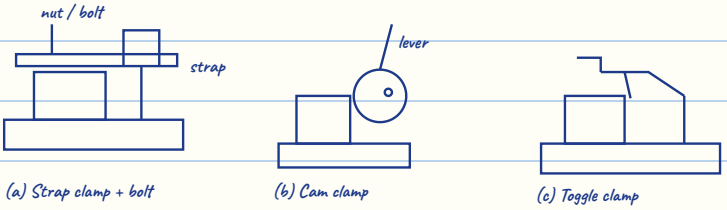


Fig 13: Strap, cam and toggle clamps

- **Strap (plate) clamp** – lever types (1st/2nd/3rd class), bolt+nut, simple, strong; slow
- **Screw clamp** – direct screw, swing/hook bolt, C-clamp, thumb screw; slow, large force
- **Cam clamp** – eccentric/spiral cam; **quick**, self-locking (angle ≤ friction), vibration may loosen; small stroke
- **Toggle clamp** – link mechanism, over-centre lock, quick, high force ratio; production/welding
- **Wedge clamp** – wedge/taper; self-locking if angle ≤ 2φ (≈ 6–8°) ; compact

22. Types of jigs

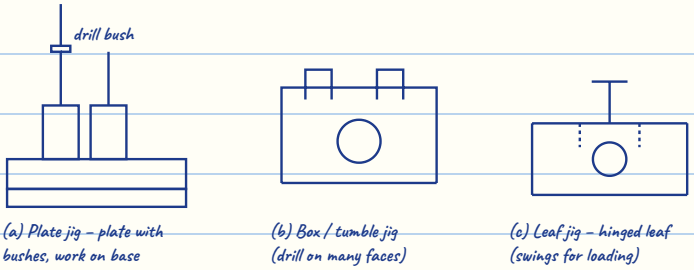


Fig 14: Plate, box and leaf jigs

- **Template jig** – simplest; plate/template laid on work, guides drill for hole pattern; not clamped; low accuracy, small batch, sheet metal
- **Plate jig** – plate with bushes + clamp; work clamped under; cheap, accurate
- **Box (tumble) jig** – box-shaped; work enclosed; rotated (tumbled) → holes on all sides without repositioning; chips need clearance
- **Channel jig** – channel section, work held by cam/screw, 1–2 sides
- **Indexing jig** – equally spaced holes on a circle; index plate + plunger
- **Diameter jig** – radial holes in round work (V-block); also trunnion/pot, sandwich jigs
- **Leaf jig** – hinged leaf (swing plate) for easy loading; small work; vibration–less rigidity

23. Types of fixtures

- **Milling fixture** – heavy; key way/tenon in T-slot; set block + feeler (tool setting); vice-type, indexing, string/gang; **straddle** milling
- **Turning (lathe) fixture** – on faceplate/chuck, balanced with counterweight; eccentric work, cranks, connecting rod
- **Grinding fixture** – magnetic/vacuum, rigid, indexing type
- **Welding fixture** – holds parts, resists distortion, toggle clamps, heat sink
- **Broaching fixture** – holds work in alignment with broach; spherical seat → self-aligning; take thrust

Bushings: press-fit (fixed), renewable, slip (liner), screw; hardened steel, clearance bush–tool 0.01–0.03; bush height ≈ 1–2 d above work.

24. Design principles checklist

★ *Important*

1. Study part drawing, operation sequence, batch size, m/c & tool, accuracy needed
2. Fix **datum** / locating surfaces (3-2-1); choose locators
3. Select **clamps** – force, direction, quick action
4. Ensure **rigidity** & strength; resist cutting force & vibration
5. **Foolproofing**: work fits only in one way; extra pins/ clearances
6. Provide **chip clearance**, easy cleaning, coolant drains
7. Easy loading/unloading, ergonomic, safe, light weight (jigs), balanced
8. Cost minimal; use standard parts; **material**: mild steel/cast iron, case-hardened locators (HRC 58–62), tool steel bushes
9. Tool guiding / setting block present; jig **feet** to rest on table

- **Advantages**: ↑ accuracy & interchangeability, ↓ time & cost, ↓ skill, ↑ safety, multi-machine operation
- **Limitations**: initial cost, design time, only economic for batch/mass production

25. Quick revision & likely exam questions

- Milling: $V = \pi DN/1000 \cdot f_m = f_z N \cdot T = (L+A+O)/f_m$
- Indexing: simple $n = 40/N$; angular $n = \theta/9$; differential ratio $= 40(A-N)/A$
- Wheel: **A 60 K 5 V** = abrasive, grain, grade, structure, bond
- Honing → bore, cross-hatch ; Lapping → loose abrasive, flat ; Broaching → one stroke, rise/tooth
- 3-2-1 = 6 DOF · Jig guides, Fixture doesn't

Probable questions

1. With neat sketch explain column & knee type milling m/c; list its parts (10M)
2. Classify milling machines. Differentiate up and down milling (10M)
3. Sketch & explain any 6 milling cutters; cutter nomenclature
4. What is indexing? Explain simple, compound, differential indexing; Find index movement for 7, 9, 28 divisions (10M)
5. Milling operations & accessories (dividing head, rotary table, vertical head)
6. Theory of grinding; self-sharpening; wheel wear
7. Explain cylindrical and centreless grinding with sketch; surface grinder types
8. Specify wheel A 60 K 5 V; abrasives & bonds; truing vs dressing; glazing & loading
9. Lapping, honing, superfinishing; compare grinding, lapping, honing
10. Broaching: types, broach nomenclature, advantages (10M)
11. Differentiate jig & fixture; explain 3-2-1 principle with sketch
12. Clamping devices; types of jigs and fixtures; design principles

UNIT IV – Mechanical & Electrical Energy Based Processes

23A03501 MACHINING PROCESSES · B.Tech Mechanical · JNTUK

I. Non-Traditional Machining (NTM) – Intro

NTM = material removed by **mechanical / thermal / chemical / electrical energy** directly – **no sharp cutting tool**, no direct tool-work hard contact.

Need for NTM

- Very hard materials (>400 HV): carbides, ceramics, HSS, Ti alloys
- Complex shapes: deep holes, micro-holes, 3D cavities, sharp corners
- Brittle / fragile jobs (glass, ceramics) – no cutting force
- Better surface finish & no burr / residual stress needed
- Tool material need NOT be harder than work

Classification (by energy source)

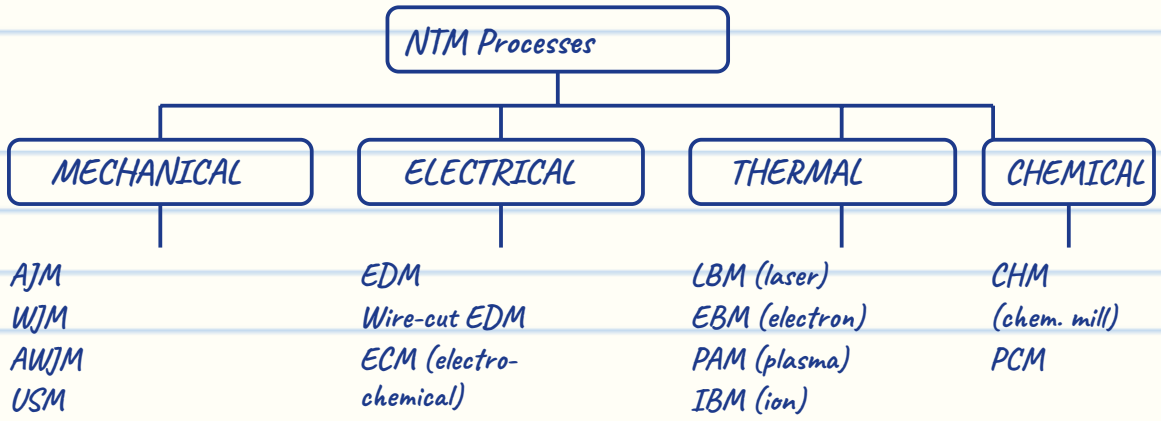


Fig 1. Classification of NTM by energy

NTM vs Conventional

Point	Conventional	NTM
Tool	Hard, sharp tool	Abrasive / jet / spark / wire
Work hardness	Limited by tool	Any hard material
Cutting force	High	Little / none
MRR	High	Low
Cost / power	Low	High
Accuracy, finish	Moderate	Very good

2. Abrasive Jet Machining (AJM)

Principle: high-velocity **gas jet ($air/N_2/CO_2$) + fine abrasive** particles strike work → **micro-chipping / erosion** (brittle fracture). Mechanical energy process.

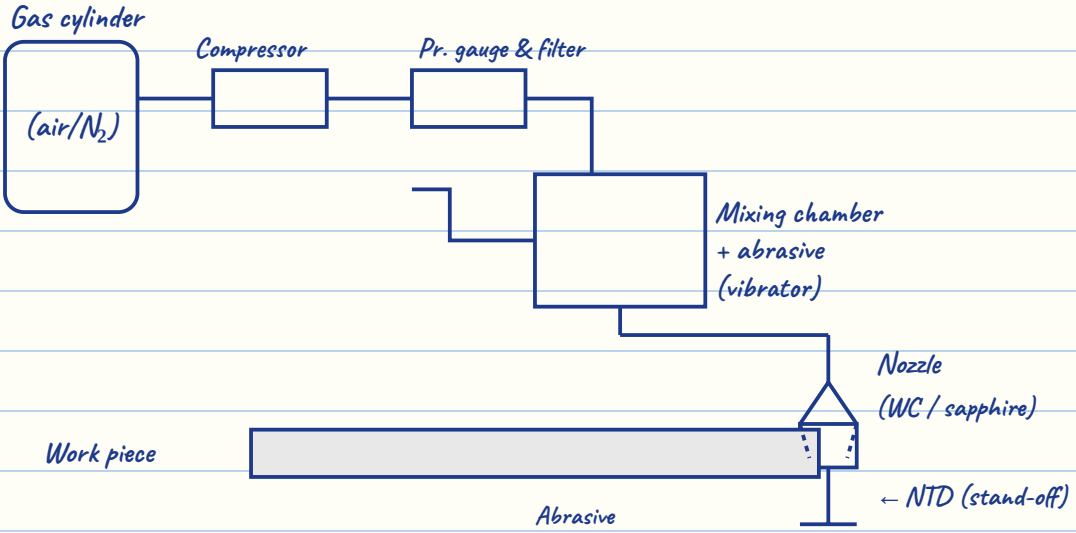


Fig 2. AJM equipment (gas → mixing chamber → nozzle → work)

Equipment

- Gas supply + compressor, pressure regulator, filter / dryer
- Mixing chamber with **vibrator** (keeps flow uniform)
- Nozzle: **WC or sapphire**, dia 0.2–0.8 mm (life 12–300 h)
- Work table / fixture, dust collector (enclosure)

Process parameters

Parameter	Typical value
Abrasive	Al_2O_3, SiC (also dolomite, glass beads)
Grain size	10–50 µm (fine); never re-use
Abrasive flow rate	3–20 g/min
Gas pressure	2–8 bar (kg/cm^2)
Jet velocity	150–300 m/s
Nozzle dia	0.2–0.8 mm
NTD / stand-off	0.25–15 mm (usually 1–2 mm)
Jet angle	60°–90°

Effect of parameters

- ↑ Pressure, ↑ flow rate → ↑ MRR (up to a limit; too much abrasive → collision loss)
- ↑ Stand-off → jet spreads → **taper, overcut** ↑, MRR first ↑ then ↓
- Finer grain → better finish, lower MRR

- Brittle work: MRR ∝ (vol per particle) ∝ $d^3.(v^2)^{(3/4)}$; ductile → low MRR

MRR model (Ghosh & Mallik – brittle, indentation)

- Each grain (dia d, velocity v, density ρ) indents hemispherical crater, width 2a
- Volume removed per impact ∝ $a^3 \propto d^3 (v\rho / H)^{(3/4)}$

$$Q = K \cdot Z \cdot d^3 \cdot (\rho v^2 / H)^{(3/4)}$$

- Z = no. of impacts/s, H = flow stress (hardness) of work, K = const
- Q ∝ abrasive flow rate (Z) → also Q ∝ v^1.5 → pressure matters

Example 1. Glass AJM: $Z = 2\times10^4$ impacts/s, volume removed per impact = 5×10^{-7} mm^3 → MRR = Z × vol = $2\times10^4 \times 5\times10^{-7}$ = **0.01 mm^3/s = 0.6 mm^3/min**. Doubling v (×2) → MRR × 2^1.5 = **2.83 times**.

Applications / Adv / Limits

- **Uses:** cutting, drilling, deburring, cleaning, etching of glass, ceramics, Ge, Si, mica; frosting; fine slots
- **Adv:** no heat damage, no vibration, can cut thin / brittle parts, low capital cost, complex shapes
- **Lim:** low MRR, **taper** in holes, abrasive embedding, nozzle wear, dust hazard, not for soft / ductile metals

3. Water Jet Machining (WJM)

Principle: **very high pressure water jet** (up to **4000 bar**) through tiny orifice → jet velocity ≈ 900 m/s (≈ 2.5 × sound speed) → **kinetic energy → high stress at work → erosion / cutting** of soft materials. (Pure water, no abrasive.)

Jet velocity (Bernoulli)

$$\tfrac{1}{2}\rho v^2 = p \Rightarrow v = \sqrt{(2p/\rho)} \; ; \; (\times Cd \text{ for losses})$$

- Eg: p = 4000 bar = 4×10^8 Pa, ρ = 1000 → $v = \sqrt{(8\times10^5)} \approx$ **894 m/s**
- Q (flow) = Cd · A · v ; Jet power = $\tfrac{1}{2}\dot{m} v^2 = p \cdot Q$

Equipment / parameters

- **Intensifier** (hydraulic oil piston, area ratio ~20:1) → 1500–4000 bar
- **Orifice (nozzle):** sapphire / ruby / **diamond**, dia 0.1–0.35 mm; life 100–200 h (diamond ~1000 h)
- Stand-off 2–6 mm; traverse speed 0.5–10 m/min
- Additives: long-chain polymers → coherent jet
- Catcher: absorbs jet energy; filters, accumulator for pulsation damping

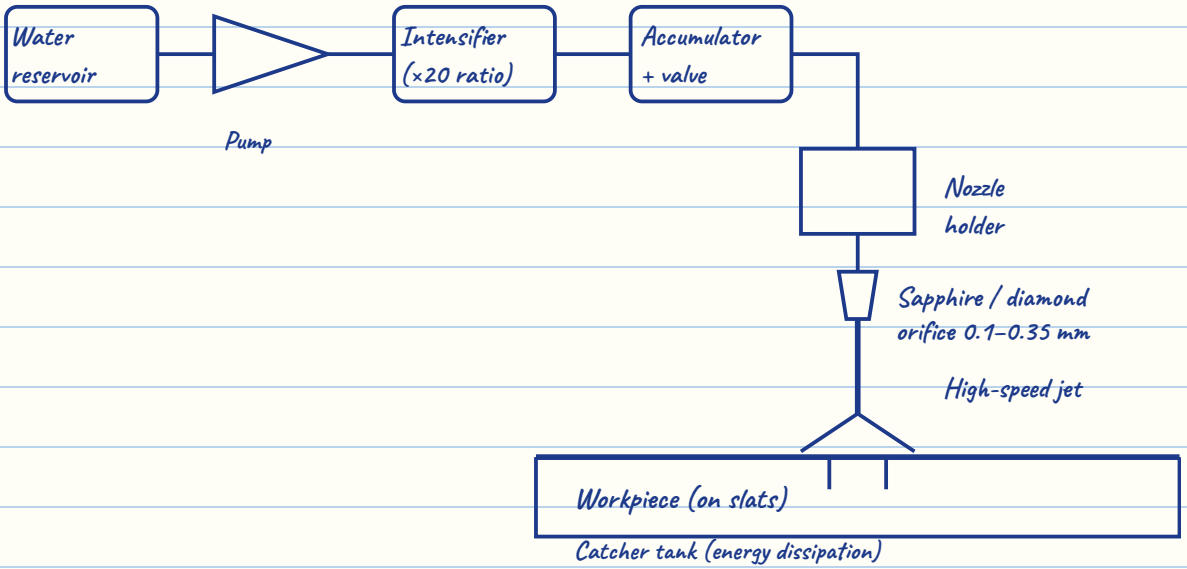


Fig 3. Water jet machine – intensifier circuit

- **Materials:** rubber, foam, paper, plastics, leather, food, wood, carpet, gasket, composites (soft)
- **Adv:** no heat (cold cut), no tool wear, narrow kerf, no dust, any contour, clean / hygienic, minimal burr
- **Lim:** cannot cut hard metals/ceramics, high capital, noise (≈ 90 dB), taper on thick jobs, kerf wanders

4. Abrasive Water Jet Machining (AWJM)

Principle: WJM + **abrasive** particles (garnet) entrained into high-pressure jet → **abrasive erosion** → cuts metals, ceramics, composites, stone. Abrasive jet speed ≈ 300–600 m/s.

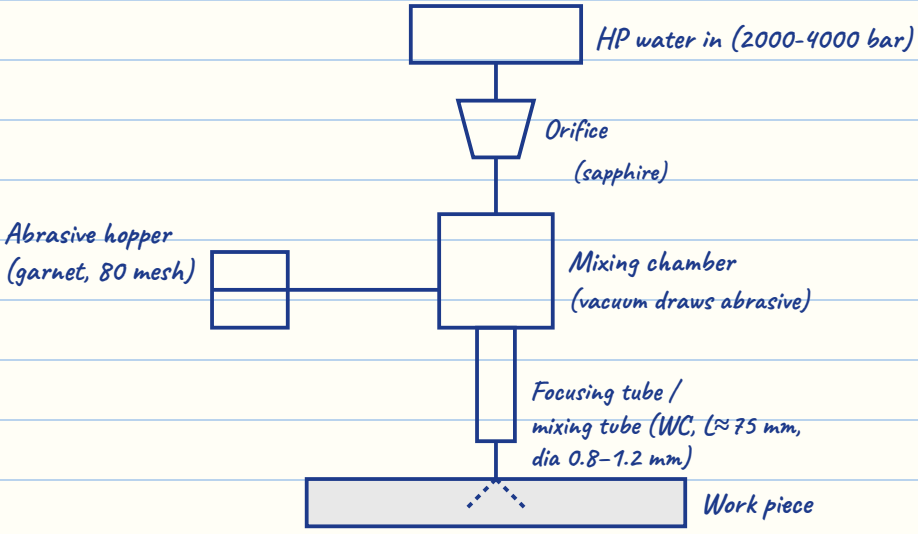


Fig 4. AWJM cutting head

- Abrasive: **garnet** (most common), Al_2O_3, olivine; mesh 60–120; flow 0.1–0.5 kg/min

- **Mixing tube** (focusing nozzle): WC / composite carbide, ID 0.76–1.2 mm, L 50–100 mm
- Orifice dia 0.25–0.4 mm; pressure 2000–4000 bar; stand-off 1–3 mm
- Mixing: **entrainment (suction type)** – common; or slurry (direct pumping) type
- Cutting speed ↑ → kerf taper, striations; ↓ speed → deeper cut
- Depth of cut $h \propto P^a \cdot \dot{m}_a^b / (V_{traverse})$ (↑ P, ↑ abrasive, ↓ speed)
- **Uses:** steel, Ti, Inconel, Al, granite, marble, glass, ceramics, CFRP; aerospace panels, tile / stone cutting
- **Adv:** no HAZ, cuts almost any material up to 200 mm thick, no thermal distortion, CNC any shape
- **Lim:** kerf taper, rough (striations), costly abrasive & disposal, nozzle (mixing tube) wear, noise, slow vs laser

WJM vs AWJM

Point	WJM	AWJM
Medium	Pure water	Water + garnet
Work	Soft, non-metals	Metals, ceramics, hard
Nozzle	Orifice only	Orifice + mixing tube
Jet dia	0.1–0.35 mm	0.8–1.2 mm

5. Ultrasonic Machining (USM)

Principle: tool vibrates at **ultrasonic frequency (≈ 20–25 kHz)**, small amplitude (15–50 µm), perpendicular to work; **abrasive slurry** between tool and work. Abrasive grains hammered → micro-chipping of brittle work. Tool shape → **mirror image** cavity.

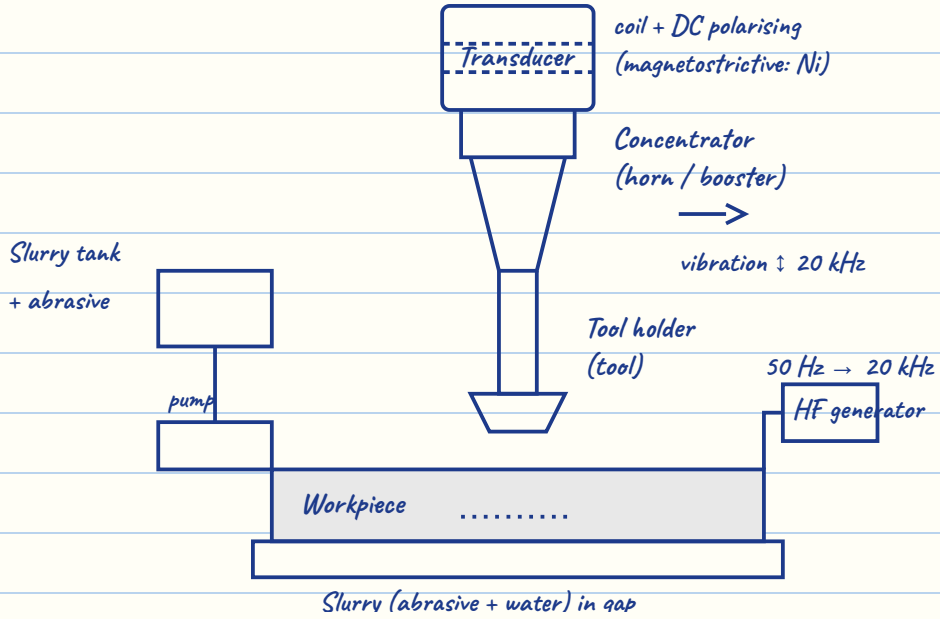


Fig 5. Ultrasonic machine

Equipment

- **HF generator:** 50 Hz mains → 20 kHz electrical signal
- **Transducer:** electrical → mechanical vibration
  - **Magnetostrictive** (Ni, Permalloy): length changes in magnetic field; ΔL = 20–30 µm; robust, efficiency ~25 %, cooling needed
  - **Piezoelectric** (quartz, PZT): efficiency 90 %, less amp., used for low power
- **Concentrator (horn):** amplifies amplitude (exponential / stepped / conical); material Monel, Ti, steel; length = $\lambda/2$ = half-wave resonance
- **Tool:** soft-tough (brass, mild steel, SS), tapered; wear ~ 1:100 (hard work)
- **Slurry:** abrasive : water ≈ 1:1 (30–40 % grit); circulated, cools & flushes debris

Horn length $L = \lambda/2 = c / 2f$ ($c = \sqrt{(E/\rho)}$)

Process parameters

Parameter	Value / effect
Frequency f	19–25 kHz (resonance)
Amplitude a	15–50 µm; MRR ↑ with a
Abrasive	B_4C (best, costly), SiC, Al_2O_3, diamond
Grit size d	#100–#800 (10–100 µm); coarse → ↑ MRR
Slurry conc. C	20–40 % by volume
Static feed force F	Optimum exists (max MRR)
Work hardness	↑ H → ↓ MRR

MRR – Shaw's model (assumptions)

- Abrasive grains spherical dia d; work brittle → crater is hemisphere, dia D (chip)

- Volume removed/grain per cycle $\propto D^3$; from geometry $D \approx 2\sqrt{(d \cdot h)}$, h = indentation depth
- So vol/grain $\propto (d \cdot h)^{(3/2)}$; N grains hit per cycle ($N \propto C/d^2$)
- f cycles/s → **$MRR = K \cdot N \cdot (d \cdot h)^{(3/2)} \cdot f$**

$$Q = K \cdot N \cdot f \cdot (a \cdot d)^{(3/2)} \Rightarrow Q \propto f \cdot a^{(3/2)} \cdot d^{(3/2)}$$

- Conclusion: MRR $\propto$ **f, $a^{(3/2)}$, $d^{(3/2)}$**, and ↓ with work hardness; Q = vol/unit time
- Typical MRR: glass ≈ 5–15 mm^3/min (depends on tool area); K absorbs work hardness H

Example 2. USM, f = 20 kHz, vol per grain-hit = 1×10^{-6} mm^3, grains hitting per cycle = 10 →
$Q = 20000 \times 10 \times 10^{-6}$ = **0.2 mm^3/s = 12 mm^3/min**. If amplitude is raised ×1.5 → $Q \times 1.5^{1.5} = 1.84$ → **0.37 $mm^3/s \approx$ 22 mm^3/min.**

Applications / Adv / Limits

- **Uses:** glass, quartz, ceramics, diamond, WC dies, Ge/Si wafers; drilling, engraving, wire-drawing dies, cutting gems, non-round holes
- **Adv:** works non-conductive brittle materials, no heat / stress, no burr, any shape hole, good finish (0.2–0.8 µm Ra)
- **Lim:** **low MRR**, high tool wear, deep holes hard (slurry access), unsuitable for soft ductile metals, expensive setup, limited size of hole

6. Comparison: AJM · WJM · AWJM · USM

Item	AJM	WJM	AWJM	USM
Medium	Gas+abrasive	Water	Water+garnet	Slurry+tool
Mechanism	Erosion	Erosion/shear	Abrasive erosion	Hammering, chipping
Pressure/Freq	2–8 bar	≤ 4000 bar	≤ 4000 bar	20 kHz
Tool	Nozzle	Sapphire orifice	Orifice+mixing tube	Shaped tool
Work	Brittle thin	Soft	Almost all	Hard brittle
MRR	Very low	High (soft)	Medium	Low
Finish	Good	Fair	Rough	Very good
HAZ	None	None	None	None

7. Electric Discharge Machining (EDM)

EDM (spark erosion / spark machining): material removed by **repeated electric sparks** between tool (electrode) & work, both **immersed in dielectric**, gap 0.01–0.5 mm. Work must be **electrically conductive**. Thermal process. (Invented by Lazarenko, 1943).

Working principle (mechanism)

1. Voltage applied across gap (60–300 V); electrode (–) work (+) usually
2. Ionisation: field ≈ 10^5–10^6 V/mm → dielectric breaks down at shortest gap
3. **Plasma channel** forms, current flows, **spark** (temp 8000–12000 °C, 10^8 W/mm^2)
4. Local melting + vaporisation of tiny crater (≈ 1–100 µm)
5. Pulse OFF → plasma collapses, dielectric flushes molten globules (debris)
6. Servo moves tool → gap kept constant; cycle repeats 2000–500000 /s

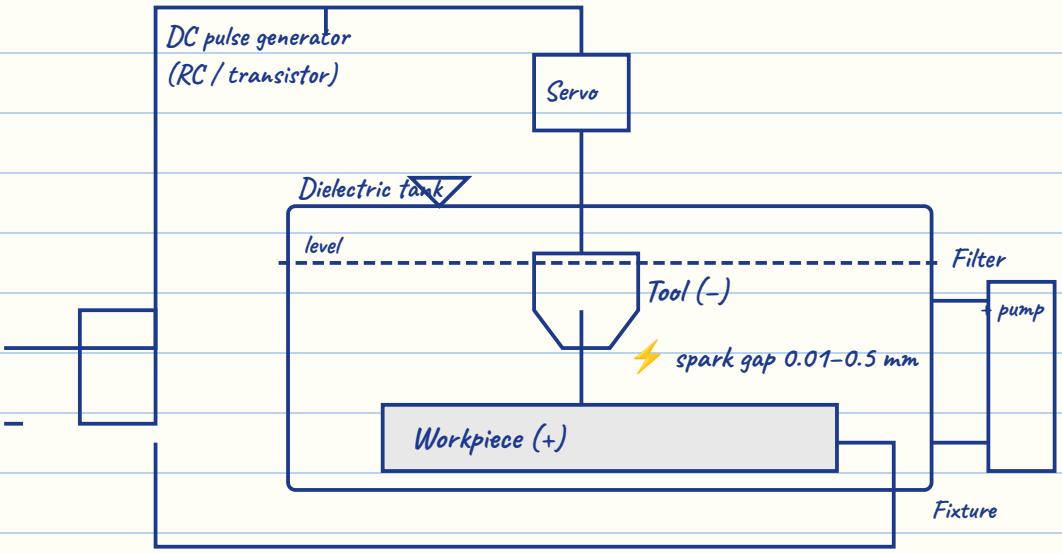


Fig 6. EDM schematic

Main elements

- **Power supply:** DC pulse (RC relaxation / transistor-pulse / RLC)
- **Servo feed system:** keeps gap constant, retracts on short-circuit
- **Dielectric system:** tank, pump, filter (≈ 5 µm), heat exchanger
- Electrode (tool) + work holder, table (X-Y)

Dielectric fluid & functions

- Types: **kerosene, EDM oil (hydrocarbon), transformer oil, deionised water** (wire-EDM), glycol
- Functions: (1) insulates till breakdown (2) **ionises** (3) **flushes** debris (4) **cools** tool/work (5) constricts spark → ↑ energy density

- Req: high dielectric strength, low viscosity, high flash point, non-toxic, cheap

Flushing methods

- **Pressure flushing** through tool hole (most common)
- **Suction flushing** (vacuum through work/tool) – less taper
- **Side / jet flushing** – nozzle directs fluid at gap
- **Pulsed / jump flushing** – tool lifts periodically; **rotating electrode**

8. RC Relaxation Circuit (Lazarenko)

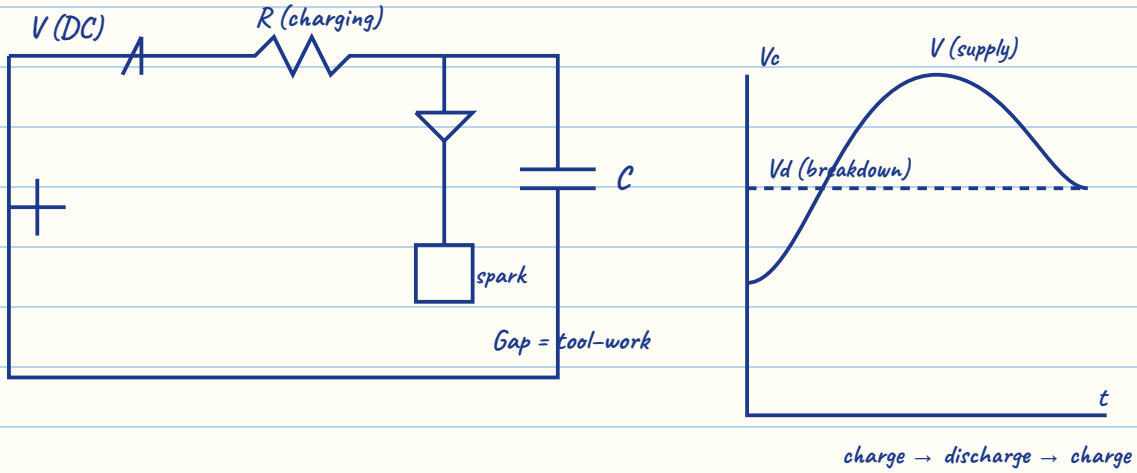


Fig 7. RC circuit & capacitor voltage cycle

- R charges C from V; when Vc = **Vd (breakdown)** → spark → C discharges → repeats
- R ≫ spark-gap resistance, so C discharges only through gap

Derivation

$$V_c = V\,(1 - e^{(-t/RC)})$$

- Time to reach Vd : $t_c = RC \cdot \ln[V / (V - V_d)]$
- Discharge time $t_d \ll t_c$, so **cycle time $T \approx t_c$**; frequency $f = 1/t_c$
- Energy per spark: $E = \frac{1}{2} C V_d^2$ → Power $P = E \cdot f = \frac{1}{2} C V_d^2 f$

$$P = \frac{1}{2} C V_d^2 / [RC \ln(V/(V-V_d))] = \frac{1}{2} V_d^2 / [R \ln(V/(V-V_d))]$$

- Let $V_d = x \cdot V$ → $P = (V^2/2R) \cdot x^2 / \ln[1/(1-x)]$
- For max P: $dP/dx = 0$ → solve gives **$x = 0.716 \approx 0.72$**

★ Important

- Max power (max MRR) when **Vd = 0.72 V** (break-down at 72 % of supply)
- MRR ∝ energy/spark × freq ; **MRR ≈ 40·I·Tm^(-1.23)** (empirical, mm^3/min, I amp, Tm melt pt °C)
- Finish ↑ with small C (low energy) but MRR ↓

Example 3. V = 200 V, R = 50 Ω, C = 10 µF, Vd = 144 V (=0.72V).

$t_c = RC \ln[V/(V-Vd)] = 50\times10\times10^{-6} \times \ln(200/56) = 5\times10^{-4} \times 1.273 =$ **0.636 ms**

$f = 1/t_c =$ **1571 Hz** ; $E = ½CVd^2 = ½\times10^{-5}\times144^2 =$ **0.1037 J**

$P = E\cdot f = 0.1037 \times 1571 \approx$ **163 W**

Example 4. Same circuit; if specific removal ≈ 0.02 mm^3 per joule (assumed for steel) → **MRR ≈ 0.02 × 163 ≈ 3.3 mm^3/s ≈ 196 mm^3/min.**

Other pulse generators

- **Transistor (controlled pulse):** square pulse, set on-time (Ton), off-time (Toff), current; modern, ↓ tool wear, high MRR
- RC: simple, slow, rough finish; used for micro-EDM, finishing

9. EDM Process Parameters

Parameter	Effect
Pulse-on time Ton (1–2000 µs)	↑ Ton → ↑ MRR, ↑ crater, ↑ Ra, ↑ HAZ, ↑ overcut
Pulse-off time Toff	↑ → better flushing, stable; ↓ MRR
Peak current Ip (1–100+ A)	↑ MRR (∝ Ip), ↑ roughness, ↑ tool wear
Gap voltage (40–100 V)	↑ gap, ↑ overcut
Open-circuit V (60–300 V)	Breakdown / gap width
Polarity	Tool –ve (work +) → max MRR; tool +ve for graphite & low wear
Flushing pressure	Debris removal, stability

Duty cycle $\eta = Ton / (Ton + Toff) \times 100\ \%$

- Pulse frequency f = 1/(Ton+Toff); higher η → higher MRR but arcing risk
- Typical: MRR 5–500 mm^3/min; Ra 0.2–12 µm; accuracy ±0.05 mm; gap 0.01–0.5 mm

Electrode (tool) materials

- **Copper** – good conductivity, fine finish, tool wear low

- **Graphite** – cheap, easy to machine, high MRR, sublimes, most used
- **Brass** – easy to make, high wear; micro-holes, **Cu-W / Ag-W** – lowest wear, costly, for carbides
- Req: high melting point, conductivity, easily machined, low wear

Tool wear, overcut, surface

- **Tool wear ratio (TWR)** = vol. tool removed / vol. work removed (graphite ~1–10 %, Cu 2–10 %, Cu-W ~ 1 %)
- Wear: corner wear > end wear > volume wear; wear ∝ 1/melting pt
- **Overcut** = (hole size – tool size)/2 = spark gap; **about 0.01–0.5 mm**; gives tool size = cavity – 2 × gap
- **Surface:** crater overlap → matt finish; Ra ↑ with Ip, Ton

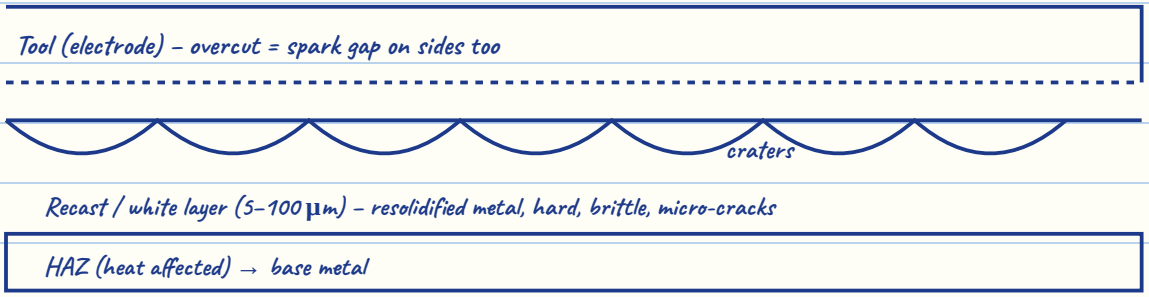


Fig 8. Machined surface: craters, recast layer, HAZ

- **Recast layer:** molten metal re-solidified; hard, micro-cracks, tensile residual stress → ↓ fatigue; removed by polishing / finishing pass
- HAZ below it: heat-affected zone with altered structure

Applications / Adv / Limits

- **Uses:** forging / extrusion / press dies, moulds, punches, turbine blades, cooling holes, deep micro-holes, tool-steel & carbide cutting, broken tap removal
- **Adv:** any conductive hard material, complex 3D cavities, no cutting force, thin delicate parts, good accuracy, burr-free, can be automated (CNC)
- **Lim:** only **conductive** work, low MRR, **tool wear**, recast layer / HAZ, sharp internal corners difficult, high power / cost, fire hazard (oil), taper

10. Wire-Cut EDM (WEDM)

Principle: same spark erosion but tool = **continuously moving thin wire** (like a band saw) – **travelling wire** cuts the work along a programmed (CNC) path in **deionised water**. No contact; wire is used once.

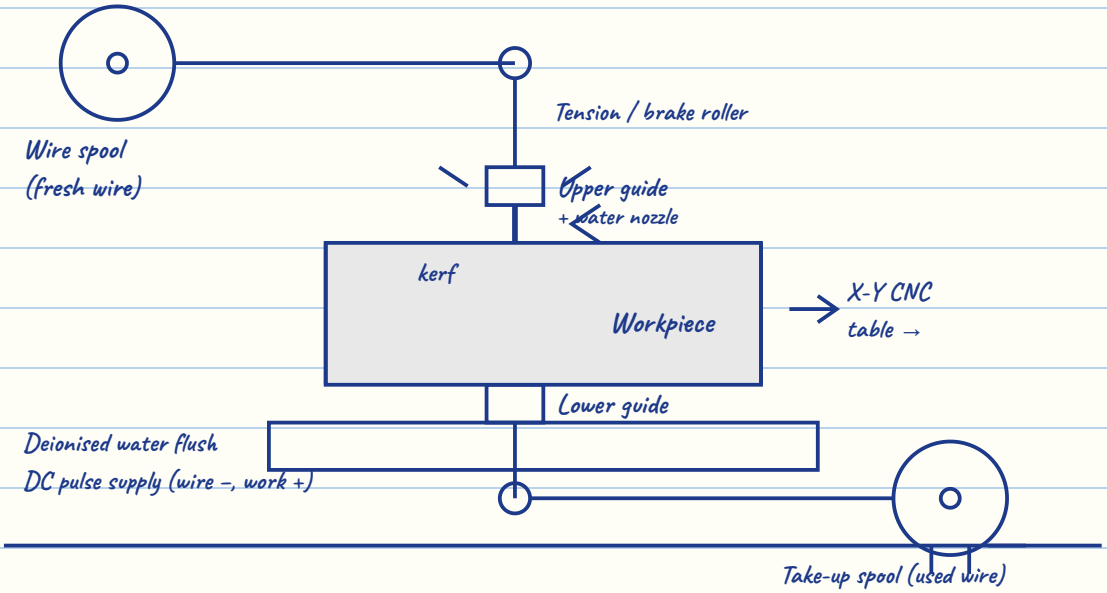


Fig 9. Wire-cut EDM

Features / parameters

- **Wire:** brass (Cu 63 / Zn 37), **zinc-coated brass**, copper, Mo, tungsten (fine); dia **0.05–0.3 mm** (0.25 usual); tensile strength ~ 900 N/mm^2; speed 1–15 m/min, used once
- **Wire guides:** sapphire / diamond / ceramic; upper & lower; give accuracy ±0.005 mm
- **Spool, tension & wire feed system:** constant tension 5–20 N; wire cassette
- **CNC X-Y table** (+ U-V upper guide for **taper cutting** up to ±30°)
- **Dielectric: deionised water** (resistivity 10^4–10^5 Ω·cm) – jets coaxial with wire, with ion-exchange resin filter
- Gap ≈ 0.025–0.05 mm; kerf = wire dia + 2×gap; Ton 0.1–25 µs; Ip 100–500 A (peak)
- Wire path programmed: **G-code**; start hole needed (or EDM drilled); "skim cuts" for finish
- Cutting speed (mm^2/min) = feed × job thickness; 100–300 mm^2/min; Ra 0.3–2.5 µm

$$MRR = \text{kerf width} \times \text{job height} \times \text{cutting speed}$$

Example 5. Wire 0.25 mm, gap 0.03 mm each side, height 40 mm, feed 3 mm/min → kerf = 0.25 + 2×0.03 = 0.31 mm → **MRR = 0.31 × 40 × 3 = 37.2 mm^3/min.**

- **Uses:** punches, dies, stripper plates, extrusion dies, gears, templates, turbine slots, medical parts, PCD / carbide tools, prototype

- **Adv:** no special tool shape (no electrode making), complex profiles, high precision, no wear (wire fresh), unattended running, tapered/ 2½D, hard materials
- **Lim:** only through (profile) cuts, conductive material, slow, high initial & running cost (wire), recast layer, thick sections slow

EDM (die-sinking) vs Wire EDM

Point	Die-sinking EDM	Wire EDM
Tool	Shaped electrode (Cu/graphite)	Moving wire (0.05–0.3 mm)
Dielectric	Kerosene / EDM oil	Deionised water
Tool wear	Significant	Negligible (fresh wire)
Shapes	Blind cavities, 3D	Through, profile cuts
Tool cost	High (making electrode)	No tool making
Flushing	Pressure / suction	Water jets coaxial
Typical use	Moulds, forging dies	Punches, dies, gears

II. Quick Revision & Likely Questions

★ *Important*

- AJM: gas 2–8 bar, 10–50 µm Al_2O_3/SiC, 150–300 m/s, nozzle WC/sapphire, NTD 1–2 mm
- WJM: ≤ 4000 bar, $v \approx \sqrt{(2p/\rho)} \approx$ 900 m/s, sapphire/diamond orifice 0.1–0.35 mm
- AWJM: + garnet; mixing (focusing) tube WC; cuts hard metals, no HAZ
- USM: 20 kHz, 15–50 µm amp.; B_4C/SiC/Al_2O_3 slurry; MRR ∝ f·a^1.5·d^1.5; transducer + horn
- EDM: spark erosion in dielectric; $E = \frac{1}{2}CV^2$; **Vd = 0.72 V** for max power; MRR↑ with Ip, Ton
- WEDM: brass wire 0.05–0.3 mm, DI water, CNC, taper cutting, no tool wear

Mnemonics

- NTM energy: **M-E-T-C** = Mechanical, Electrical, Thermal, Chemical
- EDM fluid functions: **I-I-F-C** – Insulate, Ionise, Flush, Cool
- USM chain: Generator → Transducer → Horn → Tool → Slurry → Work

Likely questions

1. Classify NTM; compare with conventional machining. (5 M)
2. With a neat sketch explain AJM; process parameters, merits & demerits. (10 M)
3. Explain WJM and AWJM; differences and applications. (10 M)
4. Describe USM with neat sketch; derive MRR (Shaw). Transducers? (10 M)
5. Principle of EDM; dielectric & flushing; electrode materials. (10 M)
6. Derive for RC circuit: f, E, and show optimum Vd = 0.72 V. (10 M)
7. Wire-cut EDM: working, parameters, applications; compare with EDM. (10 M)
8. Short notes: recast layer, overcut, tool wear, duty cycle, horn, garnet, taper cutting. (2–5 M)
9. Numericals: RC-circuit spark energy, frequency, power; MRR of USM/AJM.

UNIT V – Chemical, Electrochemical & Thermal Energy Based Processes

23A03501 MACHINING PROCESSES · B.Tech Mechanical · JNTUK

1. Chemical Machining (CHM)

CHM = controlled **chemical dissolution** (etching) of metal from selected areas using a strong **etchant**; other areas protected by **maskant**.

- No tool, no force, no heat → stress-free, burr-free surface
- Material removed by **corrosive attack** (reaction: metal + etchant → soluble salt + H_2)
- Types:
  - **Chemical milling** – shallow cavities, weight reduction (large panels)
  - **Chemical blanking** – cut thin sheet (<0.5 mm) through, intricate shapes
  - **Photochemical machining (PCM)** – maskant applied by photo-resist; used for PCBs, fine meshes
  - Also: chemical engraving, chemical deburring

Steps in chemical milling

1. **Cleaning** – remove oil, grease, scale (solvent / alkali) → good maskant adhesion
2. **Masking** – coat whole part with maskant (dip / spray / brush), cure
3. **Scribing** – cut & peel maskant from areas to be machined (template)
4. **Etching** – dip in etchant tank (temp-controlled, agitated)
5. **Rinsing & demasking** – wash, strip maskant, final clean

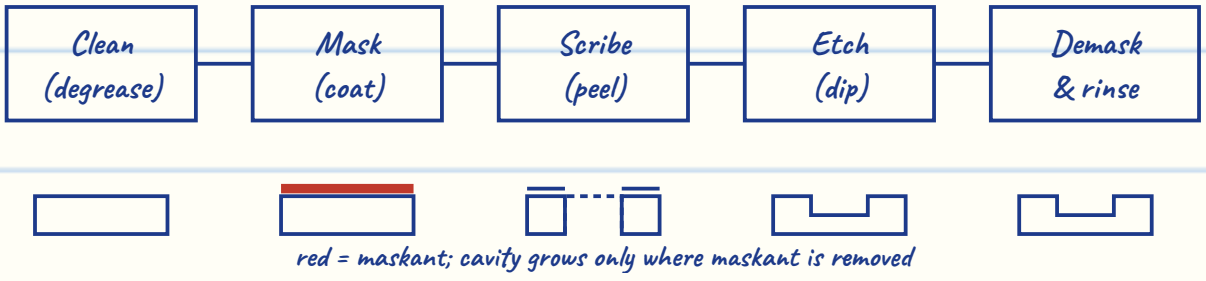


Fig 1. Sequence of chemical milling

Maskants (resists)

- Must: stick well, resist etchant, easy to scribe/strip, no reaction with metal

Maskant	Use / remark
Neoprene (rubber)	Dip/spray; most common for chem milling, 0.2–0.5 mm film; Al, Mg, Ti
PVC (vinyl)	Strippable; good for acid etchants
Polyethylene	Resist to strong acids; sheet/film form
Photoresist	Light-sensitive (KPR); fine details (±0.01 mm) → PCM

Techniques of applying maskants

- **Cut-and-peel** – dip/spray coat → cure → template placed → cut with knife → peel region by hand. Large parts, low volume, accuracy ±0.1 mm
- **Screen printing** – maskant pushed via silk/steel screen stencil; pattern printed directly; no scribing; medium accuracy, high volume (small flat parts)
- **Photoresist (photographic)** – coat with photoresist → expose UV through **negative** → develop → bake; polymerised area stays. High accuracy, fine/thin parts

Photochemical machining (PCM) steps

- Artwork (master) → photo-tool negative → coat sheet with resist → UV expose → develop → etch (spray $FeCl_3$) → strip resist

Etchants

Work material	Etchant	Remark
Aluminium	NaOH (10–20%) / $FeCl_3$	NaOH ~ 80 °C
Copper	$FeCl_3$ (also ammonium persulphate)	PCB making
Steel	HNO_3 / HCl (+ $FeCl_3$)	Fe-chloride for stainless
Titanium	HF + HNO_3 (HF 10–20%)	H_2 pick-up risk
Magnesium	H_2SO_4 / HNO_3 dilute	Fast attack

- Mnemonic: **Al–NaOH, Cu–$FeCl_3$, Ti–HF, Steel–acids**

Etch rate, undercut, etch factor

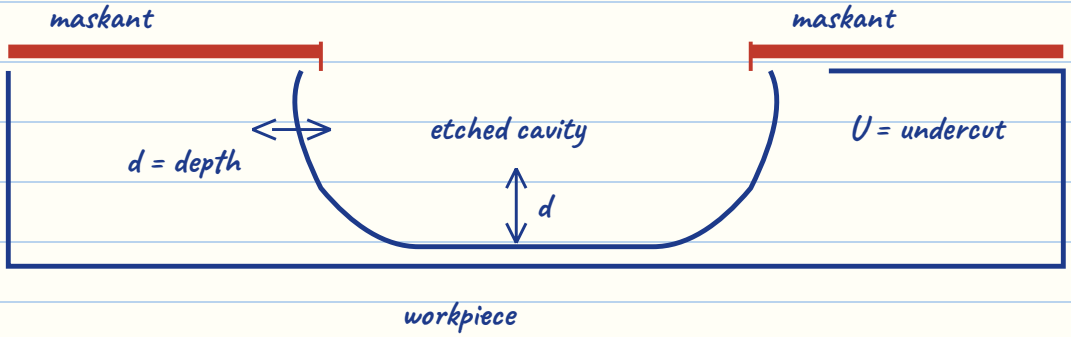


Fig 2. Undercut under the maskant (etching is isotropic)

- Etching is **isotropic** → sideways attack = **undercut (U)**

> Etch factor Fe = U / d (or d / U)

> Etch rate = depth / time (mm/min) ; U ≈ d (for Fe≈1)

- Typical etch rate 0.02–0.04 mm/min; depth 0.01–12 mm; tolerance ±0.05–0.1 mm
- Time t = d / etch rate; MRR = (area A × depth d) / t = A × etch rate
- Small etch factor (better) ⇒ sharper walls; **corner radius ≈ depth**

Process parameters

- Etchant concentration & temperature (±1–2 °C), agitation, maskant thickness, material composition/grain, surface finish (0.5–2.5 µm Ra)

Applications, advantages, limitations

Applications	Advantages	Limitations
Aircraft skin panels, wing/fuselage weight reduction; PCBs; nameplates; turbine blades; thin foils, meshes; deburring; micro-parts	No tool wear / force; no burrs, no HAZ; complex shapes; many parts at once; low capital; any hard material	Slow (min/mm); only shallow depth; undercut → low accuracy; handling of corrosive/toxic chemicals, disposal; porosity / grain attack; maskant cost; hydrogen embrittlement

2. Electrochemical Machining (ECM)

ECM = reverse of electroplating. Workpiece (**anode +**) dissolves anodically in a flowing **electrolyte** when DC passes across a small gap to the shaped **tool (cathode −)**.

- Based on **Faraday's laws of electrolysis**
- No contact, no tool wear, no heat damage; tool shape copied into work (negative copy)
- Electrolyte carries away metal hydroxide sludge & heat & H_2 gas
- Reactions (Fe in NaCl): $Fe \rightarrow Fe^{2+} + 2e^-$; $2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$; $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2\downarrow$ (sludge)

Faraday's laws

- **I law:** mass dissolved m ∝ charge Q = I·t
- **II law:** m ∝ equivalent weight E = A/Z (A = atomic wt, Z = valency)

> $m = (A \cdot I \cdot t) / (Z \cdot F)$ [F = 96500 C/mol]

Derivation of MRR

1. With current efficiency η: $m = \eta A I t / (Z F)$
2. Mass rate $\dot{m} = m/t = \eta A I / (Z F)$ (g/s)
3. Divide by density ρ → volume rate:

$$MRR = \eta A I / (\rho F Z) \ cm^3/s$$

- Also MRR = (A/ZF)(η)(J·a) with **J = current density**, a = area; and MRR = a × f (f = feed rate) ⇒ **$f = \eta AJ/(\rho F Z)$**
- Alloy: use $1/E = \Sigma(x_i \cdot Z_i/A_i)$ form: equivalent weight of alloy $E = 1/\Sigma(x_i Z_i/A_i)$, x_i = wt fraction

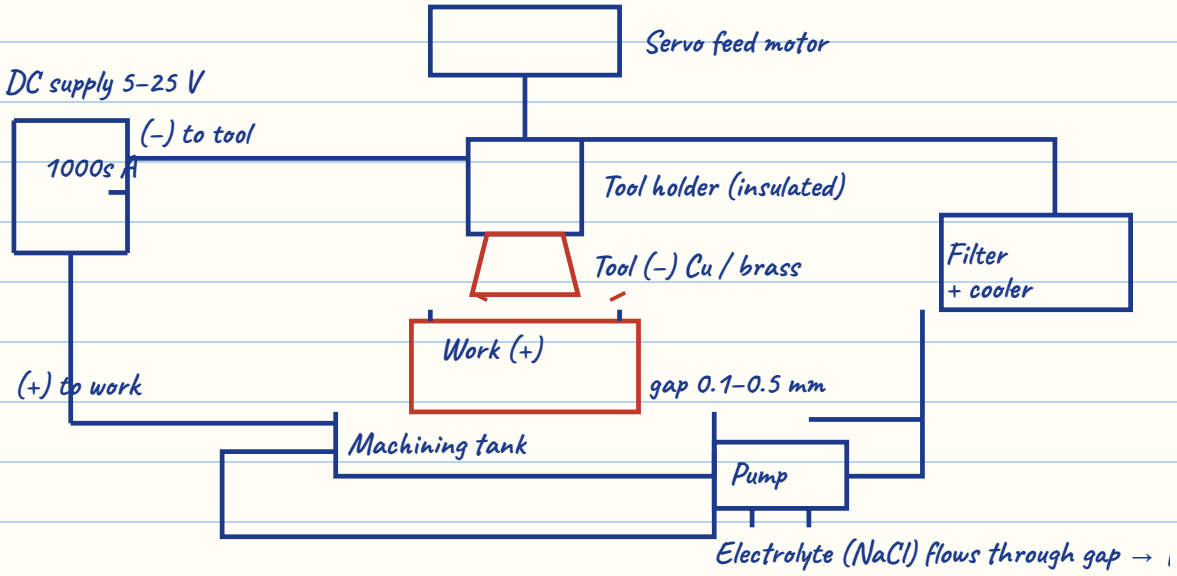


Fig 3. Schematic of ECM set-up

Equipment (main units)

- **Power supply** – low-V (5–25 V), very high-I (50–40000 A) DC, ripple-free
- **Tool feed system** – servo motor/hydraulic, constant feed 0.1–20 mm/min, short-circuit protection
- **Electrolyte system** – tank, pump (flow 10–60 m/s in gap, 1–3 MPa), filter/centrifuge, heat exchanger, pH control
- **Work-holding fixture** (insulated), machine tank, tool holder

Tool materials & tool design

- Tool material: **Copper, brass, stainless steel**, Ti, Cu-W (good conductivity, machinable, corrosion resistant, stiff)
- Tool is **never worn** (only cathode, H_2 evolves)
- Tool design aspects:
  - Cathode shape ≠ exact work shape → **overcut** compensation (tool made smaller by gap)
  - Insulate tool sides (epoxy / Teflon) → avoid stray machining

- Provide flow holes / passages for uniform electrolyte flow; avoid dead zones & sharp corners
- Rigid, avoids vibration; tool shape found by trial-and-error / CAD (cut-and-try)

Electrolytes

➤ Needs: high conductivity, non-toxic, stable, non-corrosive to machine, low cost, controls passivation

Electrolyte	Features
NaCl (10–30%)	Cheap, high conductivity, η ~100%, Fe/steel; poor accuracy (stray cut), corrosive
$NaNO_3$ (10–25%)	Passivating → better accuracy (less stray); η depends on J; costlier; used for precision parts
NaOH / KCl / $NaClO_3$	Special uses ($NaClO_3$ - precision dies)

Process parameters

Parameter	Typical value / effect
Voltage	5–25 V DC
Current density J	10–100 A/cm^2 (up to 500); MRR $\propto J$
Inter-electrode gap	0.1–0.5 mm; smaller gap ⇒ better accuracy, higher J
Feed rate	0.1–20 mm/min (typ 1–2)
Electrolyte flow/temp	~ 20–60 °C, 5–50 m/s, pressure 0.5–3 MPa
MRR	≈ 1.6 mm^3/s per 100 A (steel); surface finish 0.2–1.5 µm; accuracy ±0.02–0.1 mm

Equilibrium gap

➤ At start gap large → J low → MRR low; as gap shrinks J rises → rate rises. Gap stabilises where **dissolution speed = tool feed f** → **equilibrium gap h_e**

➤ $J = \kappa V/h$ (κ = electrolyte conductivity, V = effective voltage after drops)

$$f = \eta A \kappa V / (\rho F Z h) \Rightarrow h_e = \eta A \kappa V / (\rho F Z f)$$

➤ $h_e \propto V, \kappa$; $\propto 1/f$ → **self-regulating** process (increase f ⇒ gap decreases)

★ Important

➤ MRR = $\eta A I/(\rho F Z)$ – write derivation + units

➤ Tool wear ≈ zero; gap 0.1–0.5 mm; V 5–25 V; NaCl electrolyte

➤ ECM suits hard, tough, conductive metals – **not** insulators

Numerical (ECM)

Q. Iron ($A = 55.85$, $Z = 2$, $\rho = 7.86\ g/cm^3$) machined at $I = 1000\ A$, $\eta = 90\%$. Find MRR.

Sol. $MRR = \eta AI/(\rho FZ) = (0.9 \times 55.85 \times 1000)/(7.86 \times 96500 \times 2)$

$= 50265 / 1{,}516{,}980 =$ **$0.0331\ cm^3/s \approx 1.99\ cm^3/min$** ($\approx 33\ mm^3/s$)

Mass rate $= 0.9 \times 55.85 \times 1000/(2 \times 96500) = 0.26\ g/s$. If face area $= 2\ cm^2 \Rightarrow$ feed $f = 0.0331/2 = 0.0166\ cm/s \approx$ **10 mm/min.**

Applications	Advantages	Limitations
Turbine blades, jet-engine parts, die & mould sinking, deep holes, cavities, gun barrel rifling, deburring, contoured shapes in superalloys	High MRR for hard metals; no tool wear; no thermal damage / residual stress; burr-free; good finish; complex shapes in one pass; hardness irrelevant	Only conductive work; high power & capital cost; tool design costly; corrosion, sludge disposal; stray cutting, sharp corners hard; large floor space

3. Electro Chemical Grinding (ECG)

ECG = ECM + mechanical grinding. A **metal-bonded wheel** (conducting, with **insulating abrasive** – Al_2O_3 or diamond, CBN) is the cathode (–); work is anode (+). Electrolyte ($NaNO_3$ / NaCl) flows into the gap.

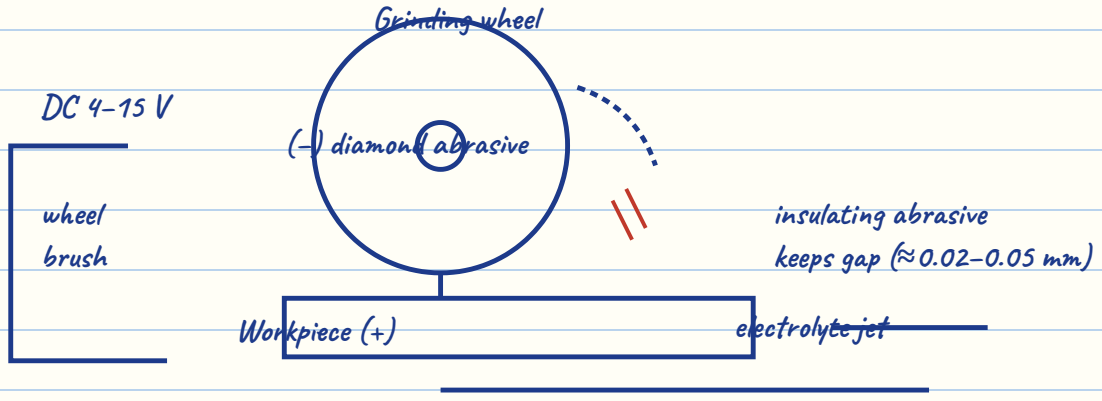


Fig 4. Electrochemical grinding

- Abrasive grains act as **insulating spacers** (gap ≈ 0.025–0.05 mm) & remove the oxide film/passive layer
- MRR split: **~90% by electrolysis, ~10% by abrasion** → wheel wear very low (up to 10× less than conv. grinding)
- Voltage 4–15 V, $J \sim 120$–$240\ A/cm^2$; wheel speed ~ 30 m/s; electrolyte $NaNO_3$ 0.25 kg/l

- Used for carbide tools, hard alloys, thin-walled/ brittle parts, burr-free sharpening
- Adv: no burr, no heat cracks, long wheel life, good finish (0.2 μm); Lim: only conducting work, corrosion, costly set-up, slower for soft metals

4. Electro Chemical Honing (ECH)

- **ECH** = ECM + honing: **honing stones** (abrasive bonded) fixed on a **cathodic tool** rotating & reciprocating in a bore; electrolyte ($NaNO_3$) flows through
- Principle: anodic dissolution (~95%) removes material; stones (low pressure) **wipe off the passive oxide film** & correct geometry
- Tool = hone with insulated metal strips + abrasive sticks; bore size ±; cycle 1–2 min
- Applications: gear teeth, cylinder bores, hydraulic sleeves, hardened steel parts

Feature	ECG	ECH	Conventional honing
Tool	Rotating wheel	Hone with stones	Hone with stones
Surface	Flat, form, external	Internal bore/gear	Bore
Removal	90% electro-chem	~95% electro-chem	100% abrasion
Stone wear	Very low	Very low	High
Heat/burr	None	None	Some heat, stresses

CHM vs ECM

Item	CHM	ECM
Principle	Chemical dissolution	Anodic dissolution (electrolysis)
Energy	Chemical, no current	Electrical DC, high current
Tool	None (maskant)	Shaped cathode
MRR	Low (0.02 mm/min)	High (1–20 mm/min)
Accuracy	±0.05–0.1 mm, undercut	±0.02–0.1 mm
Depth	Shallow	Deep cavities
Cost	Low capital	High capital
Work	Any metal	Electrically conducting

5. Laser Beam Machining (LBM)

LASER = Light Amplification by Stimulated Emission of Radiation. LBM: highly focused, monochromatic, coherent beam melts & vaporises material (power density ~10^6–10^9 W/cm^2).

Laser principle

- Atom absorbs energy → excited state (**pumping**)
- **Population inversion**: more atoms in upper (metastable) level than lower
- **Spontaneous emission** → photon; photon hits excited atom → **stimulated emission** (identical photon: same λ, phase, direction)
- Mirrors (100% + partial) in a resonator → amplification → coherent beam out

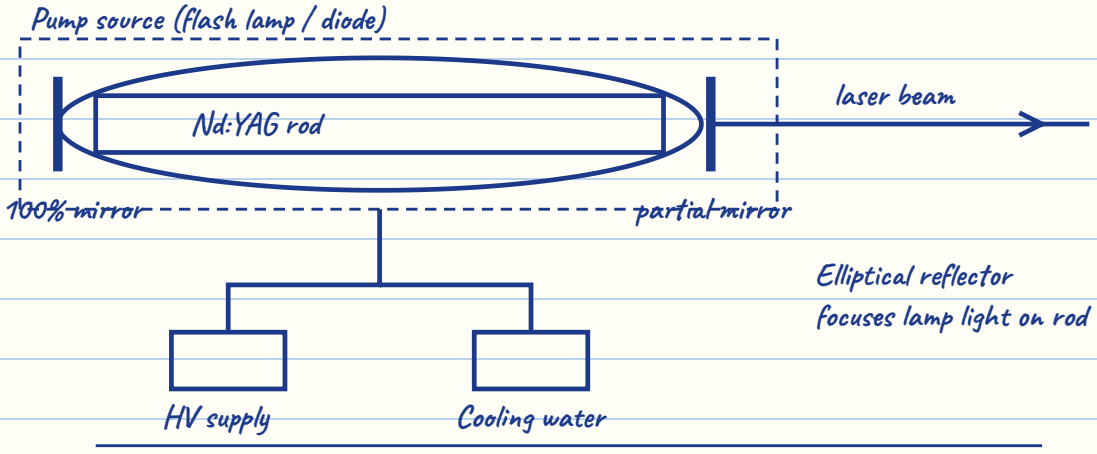


Fig 5. Nd:YAG solid-state laser

Types of lasers used

Laser	Medium / λ	Mode & power	Use
Ruby	$Al_2O_3 + Cr^{3+}$, 0.694 μm (red)	Pulsed; ~ 1 J, low eff. (<1%)	Old; spot welding, drilling
Nd:YAG	Nd^{3+} in YAG crystal, 1.064 μm (IR)	Pulsed / CW; 50 W–several kW; eff. 1–3%	Drilling, micro-welding, marking, thin cut
CO_2	$CO_2 + N_2 + He$ gas, 10.6 μm	CW/pulsed; 0.1–25 kW; eff. 10–15%	Cutting sheet, non-metals, welding

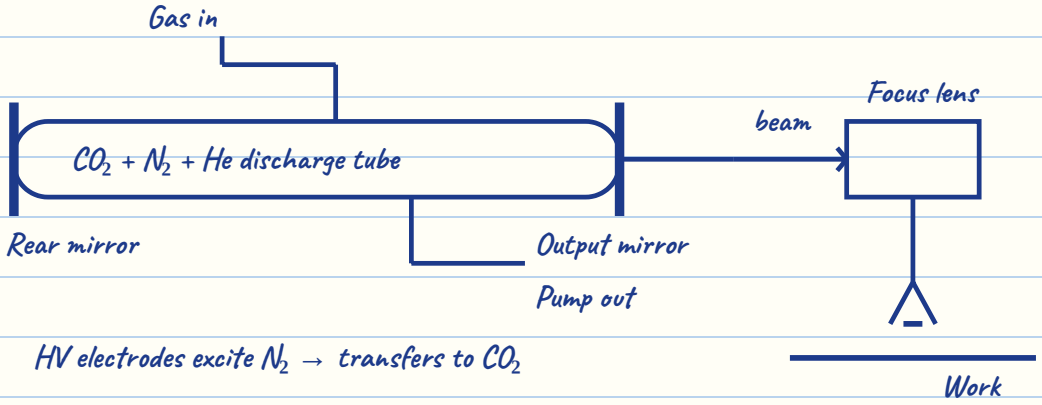


Fig 6. CO_2 laser & focusing on work (assist gas nozzle)

LBM set-up

- Laser source + power supply → beam delivery (mirrors / fibre) → focusing lens (+ assist-gas nozzle) → CNC work table. Shutter & beam monitor for control

Process parameters

- **Power / energy** (W; J per pulse): ↑ power ⇒ ↑ depth & kerf
- **Pulse duration** (ms–ns) & frequency: short pulse ⇒ less HAZ, vaporisation dominant
- **Beam spot diameter** $d = 2.44\,\lambda\, f / D$ (≈ 0.01–0.1 mm) → smaller spot, higher power density
- **Focal length** f & depth of focus: short f ⇒ small spot but shallow focus; position focus on/below surface
- **Assist gas** (O_2 – reactive, boosts cutting steel; N_2/Ar – inert, clean edge, stainless/Al; air) at 2–10 bar blows molten metal
- Work properties: reflectivity (Cu, Al reflect IR), thermal conductivity, absorptivity

$$\text{Power density} = P / (\pi\, d^2/4)\ (W/cm^2)\ ;\ \text{Energy per pulse } E = P \times t_p$$

Laser drilling & cutting

- **Drilling** – pulsed Nd:YAG; hole by melt + vaporisation + expulsion; Ø 0.01–1.0 mm, aspect ratio up to 20:1;
  - Methods: single-pulse, **percussion** (many pulses), **trepanning** (beam circles → larger holes), helical
- Applications: cooling holes in turbine blades, fuel-injector nozzles, wire-drawing dies, watch jewels, inkjet
- **Cutting** – CO_2 + assist gas; kerf 0.1–0.5 mm; steel up to 20–25 mm; speed up to m/min; CNC profiling, sheet metal, textiles, wood, plastics

- Other: welding, marking/engraving, surface hardening, micro-machining, scribing of wafers

Advantages	Limitations
Non-contact, no tool wear; any material (even diamond, ceramics); tiny precise holes; easy automation; small HAZ for short pulse	High capital & low efficiency; taper in holes; HAZ, recast layer, microcracks; depth limit (~ < 10–20 mm); reflective materials difficult; safety (eye hazard)

6. Plasma Arc Machining (PAM)

Plasma = 4th state of matter: ionised gas (ions + electrons). **PAM** – constricted high-velocity plasma jet at **10,000–30,000 K** (gas speed ~ supersonic) melts & blows away metal.

- Gas (Ar, H_2, N_2, air or mixtures) heated by arc between tungsten cathode & nozzle/work → ionises → plasma
- Gas forced through a **constricting water-cooled copper nozzle** → arc narrow, density & temp ↑
- **Gases:** Ar–H_2 (35% H_2) for thick Al/SS; N_2 (cheap) for general; air for carbon steel; Ar for inert

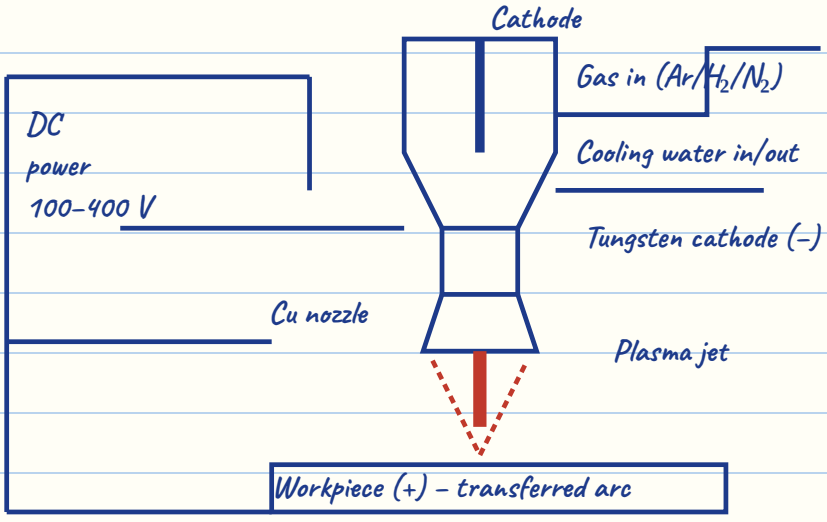


Fig 7. Plasma torch – transferred arc type

Transferred vs non-transferred arc

Item	Transferred arc	Non-transferred arc
Arc between	Electrode and **workpiece**	Electrode and **nozzle** (inside torch)
Work	Part of circuit (conductive)	Not in circuit; any material
Heat to work	High (anode heating, ~ 70% efficient)	Only plasma-jet heat (lower)
Use	Cutting, welding, metals	Spraying, non-conductors, heating

Process parameters

- Arc current 50–1000 A (typ 200–600 A), voltage 30–250 V, power 2–200 kW
- Gas type & flow (≈ 2–10 m^3/h), nozzle dia 1–5 mm, stand-off 5–10 mm
- Cutting speed (0.2–5 m/min) ↓ with thickness; torch angle; kerf 2–5 mm

Applications / HAZ

- Cutting **stainless steel, aluminium**, Cu, Ni alloys, cast iron (up to 150 mm); profile cutting of plate; shipbuilding; hard-facing; plasma spraying; turning of hard alloys (plasma-assisted)
- **HAZ**: ~ 0.5–1.5 mm — heat-affected layer + recast; taper & rough edge; may need finishing

Advantages	Limitations
Very fast cutting (≈ 10× oxy-fuel on SS/Al); cuts any conducting metal; low cost; good for thick plate; automation easy	Large kerf & HAZ; taper; metallurgical damage (cracks, oxide/nitride layer); noise, UV, fumes; poor accuracy (±1 mm); high power consumption

7. Electron Beam Machining (EBM)

EBM – a focused stream of **high-velocity electrons** (≈ 0.5–0.8 c) strikes work in **vacuum**; K.E. → heat → local melting & vaporisation. Pulsed beam drills micro-holes.

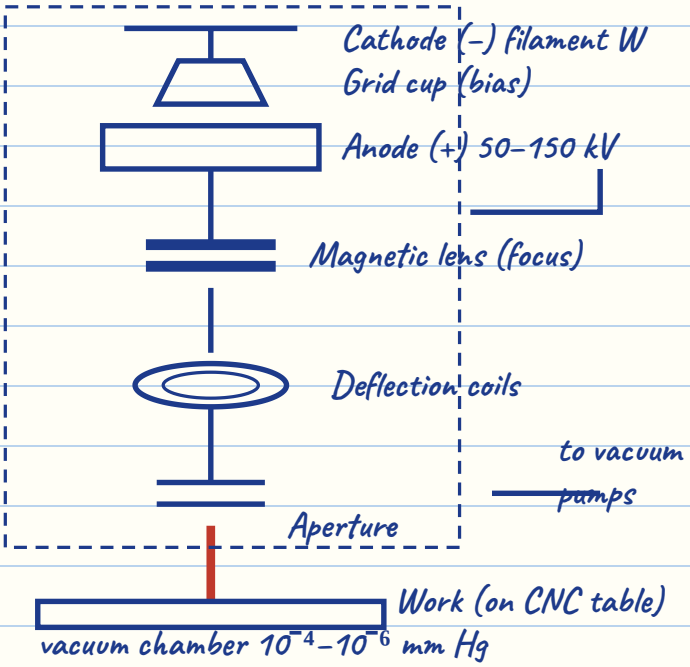


Fig 8. Electron-beam gun & column

Working

1. Tungsten filament heated → **thermionic emission**
2. Electrons accelerated by anode (**50–150 kV**) to ~ 1/2 speed of light
3. **Magnetic lens** focuses beam to Ø 10–200 µm; **deflection coils** scan/position beam
4. Power density ~ 10^{7}–10^{9} W/cm^{2} ⇒ vaporises in µs; pulses 4 µs–10 ms
5. Vacuum (≈ 10^{-5} torr) prevents scattering of electrons & oxidation of work

Process parameters

- Accelerating voltage 50–150 kV · Beam current 100 µA – 1 A (avg ~ 1–100 mA)
- Pulse duration 50 µs–10 ms, pulse frequency up to 16 kHz · Spot size 10–200 µm
- Lens current/focus, vacuum level, work material, feed (welding)

Power = V × I (e.g. 150 kV × 10 mA = 1.5 kW)

Applications

- **Micro-drilling** Ø 0.01–0.2 mm (spinnerets, filters, fuel nozzles, turbine cooling holes), slots, cutting thin foils/semiconductors, **electron-beam welding** (deep narrow weld, refractory metals), lithography (IC), annealing

Advantages	Limitations
Extremely fine holes, high precision; any material incl. diamond, W, ceramics; no mechanical force; fast (≈ 10^{4} holes/s); small HAZ; vacuum → no contamination	Very high capital cost; vacuum chamber (size limit, time to pump down); X-ray emission – shielding; skilled operator; thin sections only; recast layer

8. LBM vs PAM vs EBM

Item	LBM	PAM	EBM
Heat source	Photons (laser)	Ionised gas jet	Electrons
Medium	Air / assist gas	Ar/H_2/N_2 atmos.	Vacuum
Power density	~10^6–10^9 W/cm^2	~10^4–10^5 W/cm^2 (lowest)	Highest ~10^7–10^9 W/cm^2
Temp	~ 10,000 K local	10,000–30,000 K	~ 6000+ K local
Spot/kerf	0.01–0.5 mm	2–5 mm (large)	0.01–0.2 mm
Thickness	Thin–medium (≤ 25 mm)	Thick (≤ 150 mm)	Very thin (≤ 10 mm)
Accuracy	Good	Poor	Excellent
Work	All (non-reflective)	Conductors only	All
Cost	High	Low–medium	Very high
Main use	Drill, cut, mark	Plate cutting	Micro-holes, welding

9. NTM Process Selection Table

Process	Energy	Medium	Typical MRR	Best for
USM	Mechanical	Abrasive slurry	Low	Brittle/glass/ceramic
AJM / WJM	Mech. jet	Gas/water	Low	Cut, deburr, soft
CHM	Chemical	Etchant	Low	Shallow, sheet, large area
ECM	Electrochemical	Electrolyte	Very high	Hard conductive, cavities
EDM	Thermal (spark)	Dielectric	Medium	Dies, hard metals
LBM	Thermal	Air	Low–med	Micro-holes, cut
PAM	Thermal	Gas	Very high	Thick plate cut
EBM	Thermal	Vacuum	Low	Finest holes, welds

10. Quick Revision & Likely Questions

★ *Important*

- CHM steps: **Clean → Mask → Scribe → Etch → Demask**
- Maskants: neoprene, PVC, polyethylene, photoresist; etch factor = undercut / depth
- ECM: $MRR = \eta AI/(\rho FZ)$; gap 0.1–0.5 mm; V 5–25 V; NaCl/$NaNO_3$; Cu/brass tool, no wear
- ECG: 90% electrolysis + 10% abrasion · ECH: stones remove passive film
- Laser: stimulated emission + population inversion; Nd:YAG 1.064 μm (drill), CO_2 10.6 μm (cut)
- PAM: 10,000–30,000 K, transferred arc cuts SS/Al · EBM: 50–150 kV, vacuum, micro-holes

Likely exam questions

1. Explain chemical machining; steps, maskants, etchants, applications. (10M)
2. Define etch factor & undercut. How are maskants applied? (5M)
3. With neat sketch explain ECM; derive MRR; list advantages, limitations. (10M)
4. Discuss tool design & equilibrium gap in ECM. (5M)
5. Explain ECG and ECH; compare with conventional grinding/honing. (10M)
6. Describe laser principle & Nd:YAG / CO_2 lasers; laser drilling. (10M)
7. Explain plasma arc machining with sketch; transferred vs non-transferred. (10M)
8. Working of EBM with sketch; process parameters; applications. (10M)
9. Compare LBM, PAM, EBM; CHM vs ECM. (5M)
10. Numerical: MRR of ECM for given material, I, η.